boundary is contained in $D$. Show that the 1-cycle $\partial U = \sum \gamma_j$ is homologous to zero in $D$.

12. Let $D$ be a domain in $\mathbb{C}$ such that $\mathbb{C}^* \backslash D$ consists of $m+1$ disjoint closed connected sets. Show that there are $m$ piecewise smooth closed curves $\gamma_1, \dots, \gamma_m$ such that every integral 1-cycle $\sigma$ can be expressed uniquely in the form $\sigma = \sigma_0 + \sum k_j \gamma_j$, where the $k_j$'s are integers and $\sigma_0$ is homologous to zero in $D$. *Remark.* The $\gamma_j$'s form a **homology basis** for $D$.

# IX

# The Schwarz Lemma and Hyperbolic Geometry

This short chapter is devoted to the Schwarz lemma, which is a simple consequence of the power series expansion and the maximum principle. The Schwarz lemma is proved in Section 1, and it is used in Section 2 to determine the conformal self-maps of the unit disk. In Section 2 we formulate the Schwarz lemma to be invariant under the conformal self-maps of the unit disk, thereby obtaining Pick's lemma. This leads in Section 3 to the hyperbolic metric and hyperbolic geometry of the unit disk.

## 1. The Schwarz Lemma

The Schwarz lemma is easy to prove, yet it has far-reaching consequences.

**Theorem (Schwarz Lemma).** *Let $f(z)$ be analytic for $|z| < 1$. Suppose $|f(z)| \le 1$ for all $|z| < 1$, and $f(0) = 0$. Then*

$$|f(z)| \le |z|, \qquad |z| < 1. \tag{1.1}$$

*Further, if equality holds in (1.1) at some point $z_0 \ne 0$, then $f(z) = \lambda z$ for some constant $\lambda$ of unit modulus.*

For the proof, we factor $f(z) = zg(z)$, where $g(z)$ is analytic, and we apply the maximum principle to $g(z)$. Let $r < 1$. If $|z| = r$, then $|g(z)| = |f(z)|/r \le 1/r$. By the maximum principle, $|g(z)| \le 1/r$ for all $z$ satisfying $|z| \le r$. If we let $r \to 1$, we obtain $|g(z)| \le 1$ for all $|z| < 1$. This yields (1.1). If $|f(z_0)| = |z_0|$ for some $z_0 \ne 0$, then $|g(z_0)| = 1$, and by the strict maximum principle, $g(z)$ is constant, say $g(z) = \lambda$. Then $f(z) = \lambda z$.

An analogous estimate holds in any disk. If $f(z)$ is analytic for $|z - z_0| < R$, $|f(z)| \le M$, and $f(z_0) = 0$, then

$$|f(z)| \le \frac{M}{R}|z - z_0|, \qquad |z - z_0| < R, \tag{1.2}$$

with equality only when $f(z)$ is a multiple of $z - z_0$. This can be proved directly, based on the factorization $f(z) = (z - z_0)g(z)$. It can also be obtained from (1.1) by scaling in both the $z$-variable and the $w$-variable, $w = f(z)$, and by translating the center of the disk to $z_0$, as follows. The change of variable $\zeta \mapsto R\zeta + z_0$ maps the unit disk $\{|\zeta| < 1\}$ onto the disk $\{|z - z_0| < R\}$. If we define $h(\zeta) = f(R\zeta + z_0)/M$, then $h(\zeta)$ is analytic on the open unit disk and satisfies $|h(\zeta)| \leq 1$ and $h(0) = 0$. The estimate $|h(\zeta)| \leq |\zeta|$ becomes (1.2).

The Schwarz lemma gives an explicit estimate for the "modulus of continuity" of an analytic function. It shows that a uniformly bounded family of analytic functions is "equicontinuous" at each point. We will return in Chapter XI to treat the ideas of equicontinuity and compactness for families of analytic functions.

There is an infinitesimal version of the Schwarz lemma.

**Theorem.** *Let $f(z)$ be analytic for $|z| < 1$. If $|f(z)| \leq 1$ for $|z| < 1$, and $f(0) = 0$, then*

$$|f'(0)| \leq 1, \tag{1.3}$$

*with equality if and only if $f(z) = \lambda z$ for some constant $\lambda$ with $|\lambda| = 1$.*

The estimate (1.3) follows by taking $z \to 0$ in the Schwarz lemma. For the case of equality, we consider the factorization $f(z) = zg(z)$ used in the proof of the Schwarz lemma, and we observe that $g(0) = f'(0)$. If $|f'(0)| = 1$, we then have $|g(0)| = 1$, and we conclude as before from the strict maximum principle that $g(z)$ is constant. Hence $f(z) = \lambda z$.

Note that the estimate (1.3) is the same as the Cauchy estimate for $f'(0)$ derived in Section IV.4, without the hypothesis that $f(0) = 0$. See also Exercise 7.

## Exercises for IX.1

1. Let $f(z)$ be analytic and satisfy $|f(z)| \leq M$ for $|z - z_0| < R$. Show that if $f(z)$ has a zero of order $m$ at $z_0$, then

$$|f(z)| \leq \frac{M}{R^m}|z - z_0|^m, \qquad |z - z_0| < R.$$

Show that equality holds at some point $z \neq z_0$ only when $f(z)$ is a constant multiple of $(z - z_0)^m$.

2. Suppose that $f(z)$ is analytic and satisfies $|f(z)| \leq 1$ for $|z| < 1$. Show that if $f(z)$ has a zero of order $m$ at $z_0$, then $|z_0|^m \geq |f(0)|$. *Hint.* Let $\psi(z) = (z - z_0)/(1 - \bar{z}_0 z)$, which is a fractional linear transformation mapping the unit disk onto itself, and show that $|f(z)| \leq |\psi(z)|^m$.

3. Suppose that $f(z)$ is analytic for $|z| \leq 1$, and suppose that $1 < |f(z)| < M$ for $|z| = 1$, while $f(0) = 1$. Show that $f(z)$ has a zero in the unit disk, and that any such zero $z_0$ satisfies $|z_0| > 1/M$. *Hint.* For the second assertion, consider $\psi(f(z))$, where $\psi(w)$ is a fractional linear transformation mapping 1 to 0 and the circle $\{|w| = M\}$ to the unit circle. Or use Exercise 2.

4. Suppose that $f(z)$ is analytic for $|z| < 1$ and satisfies $f(0) = 0$ and $\operatorname{Re} f(z) < 1$. (a) Show that $|f(z)| \leq 2|z|/(1-|z|)$. *Hint.* Consider the composition of $f(z)$ and the fractional linear transformation mapping the half-plane $\{\operatorname{Re} w < 1\}$ onto the unit disk. (b) Show that $|f'(0)| \leq 2$. (c) For fixed $z_0$ with $0 < |z_0| < 1$, determine for which functions $f(z)$ there is equality in (a). (d) Determine for which functions $f(z)$ there is equality in (b). (e) By scaling the estimates in (a) and (b), obtain sharp estimates for $|g(z)|$ and $|g'(0)|$ for functions $g(z)$ analytic for $|z| < R$ and satisfying $g(0) = 0$ and $\operatorname{Re} g(z) < C$.

5. Suppose that $f(z)$ is analytic and satisfies $|f(z)| \leq 1$ for $|z| < 1$. Show that if $|f(0)| \geq r$, then $|f(z)| \geq (r-|z|)/(1-r|z|)$ for $|z| < r$. Determine for which functions $f(z)$ equality holds at some point $z_0$ with $|z_0| < r$.

6. Let $f(z)$ be a conformal map of the open unit disk onto a domain $D$. Show that the distance from $f(0)$ to the boundary of $D$ is estimated by $\operatorname{dist}(f(0), \partial D) \leq |f'(0)|$.

7. Suppose that $f(z) = \sum_{k=0}^{\infty} a_k z^k$ is analytic for $|z| < 1$ and satisfies $|f(z)| \leq M$.
   (a) Show that $\sum_{k=0}^{\infty} |a_k|^2 \leq M^2$. *Hint.* Integrate $|f(z)|^2$ around a circle of radius $r$.
   (b) Show using (a) that $|f'(0)| \leq M$, with equality only if $f(z)$ is a constant multiple of $z$. *Remark.* It is not assumed that $f(0) = 0$.
   (c) Show that $|f^{(k)}(0)| \leq k!M$, with equality only if $f(z)$ is a constant multiple of $z^k$.

8. Suppose that $f(z)$ is analytic for $|z| < 1$ and satisfies $|f(z)| < 1$, $f(0) = 0$, and $|f'(0)| < 1$. Let $r < 1$. Show that there is a constant $c < 1$ such that $|f(z)| \leq c|z|$ for $|z| \leq r$. Show that the $n$th iterate $f_n(z) = f(f(\cdots f(z)\cdots)) = f(f_{n-1}(z))$ of $f(z)$ satisfies $|f_n(z)| \leq c^n|z|$ for $|z| \leq r$. Deduce that $f_n(z)$ converges to zero normally on the open unit disk $\mathbb{D}$.

## 2. Conformal Self-Maps of the Unit Disk

We denote by $\mathbb{D}$ the open unit disk in the complex plane, $\mathbb{D} = \{|z| < 1\}$. A **conformal self-map of the unit disk** is an analytic function from $\mathbb{D}$ to itself that is one-to-one and onto. The composition of two conformal self-maps is again a conformal self-map, and the inverse of a conformal self-map is a conformal self-map. The conformal self-maps form what is called a "group," with composition as the group operation. The group identity is the identity map $g(z) = z$.

For fixed angle $\varphi$, the rotation $z \mapsto e^{i\varphi} z$ is a conformal self-map of $\mathbb{D}$ that fixes the origin, and these are the only conformal self-maps that leave 0 fixed.

**Lemma.** *If $g(z)$ is a conformal self-map of the unit disk $\mathbb{D}$ such that $g(0) = 0$, then $g(z)$ is a rotation, that is, $g(z) = e^{i\varphi} z$ for some fixed $\varphi$, $0 \leq \varphi \leq 2\pi$.*

To see this, we apply the Schwarz lemma to $g(z)$ and to its inverse. Since $g(0) = 0$ and $|g(z)| < 1$, the Schwarz lemma applies, and $|g(z)| \leq |z|$. If we apply the Schwarz lemma also to $g^{-1}(w)$, we obtain $|g^{-1}(w)| \leq |w|$, which for $w = g(z)$ becomes $|z| \leq |g(z)|$. Thus $|g(z)| = |z|$. Since $g(z)/z$ has constant modulus, it is constant. Hence $g(z) = \lambda z$ for a unimodular constant $\lambda$.

**Theorem.** *The conformal self-maps of the open unit disk $\mathbb{D}$ are precisely the fractional linear transformations of the form*

$$f(z) = e^{i\varphi} \frac{z - a}{1 - \bar{a}z}, \qquad |z| < 1, \tag{2.1}$$

*where $a$ is complex, $|a| < 1$, and $0 \leq \varphi \leq 2\pi$.*

Define $g(z) = (z - a)/(1 - \bar{a}z)$. Since $g(z)$ is a fractional linear transformation, it is a conformal self-map of the extended complex plane, and it maps circles to circles. From

$$|e^{i\theta} - a| = |e^{-i\theta} - \bar{a}| = |1 - \bar{a}e^{i\theta}|, \qquad 0 \leq \theta \leq 2\pi,$$

we see that $|g(z)| = 1$ for $z = e^{i\theta}$, so that $g(z)$ maps the unit circle to itself. Since $g(a) = 0$, $g(z)$ must map the open unit disk to itself. Consequently, $g(z)$ is a conformal self-map of the unit disk, and so is $f(z)$ defined by (2.1). Let $h(z)$ be an arbitrary conformal self-map of $\mathbb{D}$, and set $a = h^{-1}(0)$. Then $h \circ g^{-1}$ is a conformal self-map of $\mathbb{D}$, and $(h \circ g^{-1})(0) = h(a) = 0$. By the lemma, $(h \circ g^{-1})(w) = e^{i\varphi} w$ for some fixed $\varphi$, $0 \leq \varphi \leq 2\pi$. Writing $w = g(z)$, we obtain $h(z) = e^{i\varphi} g(z)$, and $h(z)$ has the form (2.1).

The parameters $a$ and $e^{i\varphi}$ are uniquely determined by the conformal self-map $f(z)$ of $\mathbb{D}$. The parameter $a$ is $f^{-1}(0)$, and since

$$f'(z) \;=\; e^{i\varphi}\frac{1-|a|^2}{(1-\bar{a}z)^2}, \qquad |z|<1,$$

the parameter $\varphi$ is uniquely specified (modulo $2\pi$) as the argument of $f'(0)$. Thus there is a one-to-one correspondence between points of the parameter space $\mathbb{D}\times\partial\mathbb{D}$ and conformal self-maps of the open unit disk.

Next we take a giant step by proving a form of the Schwarz lemma that is invariant under conformal self-maps of the open unit disk.

**Theorem (Pick's Lemma).** *If $f(z)$ is analytic and satisfies $|f(z)|<1$ for $|z|<1$, then*

$$(2.2) \qquad |f'(z)| \;\le\; \frac{1-|f(z)|^2}{1-|z|^2}, \qquad |z|<1.$$

*If $f(z)$ is a conformal self-map of $\mathbb{D}$, then equality holds in (2.2); otherwise, there is strict inequality for all $|z|<1$.*

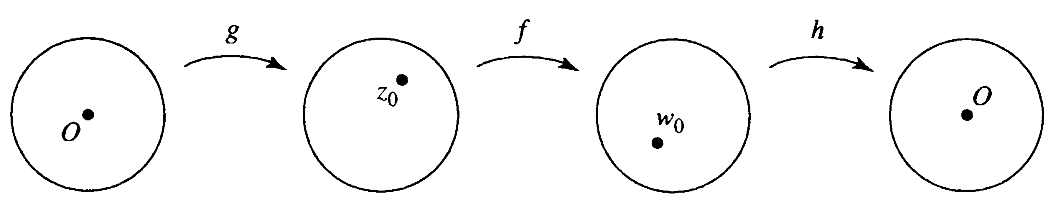
To prove (2.2), our strategy is to transport $z$ and $f(z)$ to 0 using conformal self-maps, and to apply the Schwarz lemma to the resulting composition. Fix $z_0\in\mathbb{D}$ and set $w_0=f(z_0)$. Let $g(z)$ and $h(z)$ be conformal self-maps of $\mathbb{D}$ mapping 0 to $z_0$ and $w_0$ to 0, respectively, say

$$g(z) \;=\; \frac{z+z_0}{1+\overline{z_0}z}, \qquad\qquad h(w) \;=\; \frac{w-w_0}{1-\overline{w_0}w}.$$

Then $h\circ f\circ g$ maps 0 to 0. The estimate (1.3) and the chain rule yield

$$(2.3) \qquad |(h\circ f\circ g)'(0)| \;=\; |h'(w_0)f'(z_0)g'(0)| \;\le\; 1,$$

hence $|f'(z_0)| \le 1/|g'(0)||h'(w_0)|$. Substituting $g'(0)=1-|z_0|^2$ and $h'(w_0)=1/(1-|w_0|^2)$, we obtain (2.2).




If $f(z)$ is a conformal self-map of $\mathbb{D}$, then so is $h\circ f\circ g$, so we have equality in (2.3), which yields equality in (2.2). Conversely, suppose that $f(z)$ is an analytic function from $\mathbb{D}$ to $\mathbb{D}$ such that equality holds in (2.2) at one point $z_0$. Then the calculations above give $|(h\circ f\circ g)'(0)|=1$. According to Section 1, $h\circ f\circ g$ is multiplication by a unimodular constant, hence a conformal self-map of $\mathbb{D}$. Composing by $h^{-1}$ on the left and by $g^{-1}$ on the right, we conclude that $f$ is a conformal self-map of $\mathbb{D}$.

## Exercises for IX.2

1. A **finite Blaschke product** is a rational function of the form

$$B(z) \;=\; e^{i\varphi}\left(\frac{z-a_1}{1-\overline{a_1}z}\right)\cdots\left(\frac{z-a_n}{1-\overline{a_n}z}\right),$$

where $a_1,\ldots,a_n \in \mathbb{D}$ and $0 \le \varphi \le 2\pi$. Show that if $f(z)$ is continuous for $|z| \le 1$ and analytic for $|z| < 1$, and if $|f(z)| = 1$ for $|z| = 1$, then $f(z)$ is a finite Blaschke product.

2. Show that $f(z) = (3+z^2)/(1+3z^2)$ is a finite Blaschke product.

3. Suppose $f(z)$ is analytic for $|z| < 3$. If $|f(z)| \le 1$, and $f(\pm i) = f(\pm 1) = 0$, what is the maximum value of $|f(0)|$? For which functions is the maximum attained?

4. For fixed $z_0, z_1 \in \mathbb{D}$, find the maximum value of $|f(z_1) - f(z_0)|$ among all analytic functions $f(z)$ on the open unit disk $\mathbb{D}$ satisfying $|f(z)| < 1$. Determine for which such functions the maximum value is attained. *Hint.* Consider first the case where $z_0 = r > 0$ and $z_1 = -r$, and show that the maximum is $2r$, attained only for $f(z) = \lambda z$, $|\lambda| = 1$.

5. Show that any conformal self-map of the upper half-plane has the form

$$f(z) = \frac{az+b}{cz+d}, \qquad \operatorname{Im} z > 0,$$

where $a, b, c, d$ are real numbers satisfying $ad - bc = 1$. When do two such coefficient choices for $a, b, c, d$ determine the same conformal self-map of the upper half-plane?

6. Show that the conformal maps of the upper half-plane onto the open unit disk are of the form

$$f(z) = e^{i\varphi}\frac{z-a}{z-\bar{a}}, \qquad \operatorname{Im} a > 0,\ \ 0 \le \varphi \le 2\pi.$$

Show that $a$ and $e^{i\varphi}$ are uniquely determined by the conformal map.

7. Show that every conformal self-map of the complex plane $\mathbb{C}$ has the form $f(z) = az + b$, where $a \ne 0$. *Hint.* The isolated singularity of $f(z)$ at $\infty$ must be a simple pole.

8. Show that every conformal self-map of the Riemann sphere $\mathbb{C}^*$ is given by a fractional linear transformation.

9. Show that any conformal self-map of the punctured unit disk $\{0 < |z| < 1\}$ is a rotation $z \mapsto e^{i\varphi}z$.

10. Show that any conformal self-map of the punctured complex plane $\{0 < |z| < \infty\}$ is either a multiplication $z \mapsto az$, or such a multiplication followed by the inversion $z \mapsto 1/z$.

11. Let $D = \mathbb{C} \backslash \{a_1, \ldots, a_m\}$ be the complex plane with $m$ punctures. Show that any conformal self-map of $D$ is a fractional linear transformation that permutes $\{a_1, \ldots, a_m, \infty\}$.

12. Determine the conformal self-maps of the following domains $D$: (a) $D = \mathbb{C} \backslash \{0, 1\}$, (b) $D = \mathbb{C} \backslash \{-1, 0, 1\}$, (c) $D = \mathbb{C} \backslash \{-1, 0, 2\}$.

13. Suppose $f(z)$ is an analytic function from the open unit disk $\mathbb{D}$ to itself that is not the identity map $z$. Show that $f(z)$ has at most one fixed point in $\mathbb{D}$. *Hint.* Make a change of variable with a conformal self-map of $\mathbb{D}$ to place the fixed point at 0.

14. Suppose $f(z)$ is an analytic function from the open unit disk $\mathbb{D}$ to itself that is not a conformal self-map, and denote by $f_n(z)$ the $n$th iterate of $f(z)$. Show that if $f(z)$ has a fixed point $z_0 \in \mathbb{D}$, then $f_n(z)$ converges to $z_0$ for each $z \in \mathbb{D}$. Show that for each $r < 1$, the convergence is uniform for $|z| \leq r$. *Hint.* See Exercise 1.8.

15. We say that two conformal self-maps $f$ and $g$ of $\mathbb{D}$ are **conjugate** if there is a conformal self-map $h$ of $\mathbb{D}$ such that $g = h \circ f \circ h^{-1}$. (See the exercises for Section II.7.) Let $f$ be a conformal self-map of $\mathbb{D}$ that is not the identity map $z$. (a) Show that either $f$ has two fixed points on $\partial\mathbb{D}$, counting multiplicity, or $f$ has one fixed point in $\mathbb{D}$. (b) Show that $f$ has a fixed point in $\mathbb{D}$ if and only if $f$ is conjugate to a rotation $g(z) = e^{i\varphi}z$. (c) Show that rotations by different angles are not conjugate. (d) Show that $f$ has two distinct fixed points on $\partial\mathbb{D}$ if and only if $f$ is conjugate to $g(z) = (z - s)/(1 - sz)$ for some $s$ satisfying $0 < s < 1$. (e) Show that $g$'s for different $s$'s are not conjugate. (f) Show that any two conformal self-maps of $\mathbb{D}$ with one fixed point on $\partial\mathbb{D}$ (of multiplicity two) are conjugate.

## 3. Hyperbolic Geometry

Suppose $w = f(z)$ is a conformal self-map of the open unit disk $\mathbb{D}$. From Pick's lemma we then have equality in (2.2),

$$\left|\frac{dw}{dz}\right| = \frac{1 - |w|^2}{1 - |z|^2}.$$

In differential form this becomes

$$\frac{|dw|}{1 - |w|^2} = \frac{|dz|}{1 - |z|^2},$$

which means that if $\gamma$ is any smooth curve in $\mathbb{D}$, and $w = f(z)$ is a conformal self-map of $\mathbb{D}$, then

$$(3.1) \qquad \int_{f\circ\gamma} \frac{|dw|}{1-|w|^2} = \int_\gamma \frac{|dz|}{1-|z|^2}.$$

Thus to obtain a length function that is invariant under conformal self-maps of $\mathbb{D}$, we are led to make the following definition. We define the **length of $\gamma$ in the hyperbolic metric** by

$$(3.2) \qquad \text{hyperbolic length of } \gamma = 2\int_\gamma \frac{|dz|}{1-|z|^2}.$$

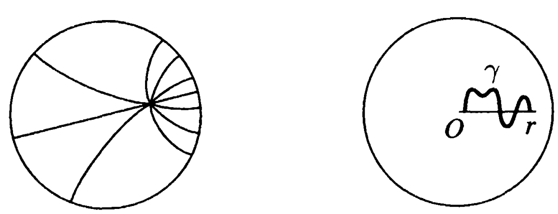
The factor 2 is a harmless factor, which is often omitted. (It adjusts the metric so that its curvature is $-1$.) The identity (3.1) shows that $f\circ\gamma$ has the same hyperbolic length as $\gamma$ for any conformal self-map $f(z)$ of $\mathbb{D}$. Thus hyperbolic lengths are invariant under conformal self-maps of $\mathbb{D}$.

We define the **hyperbolic distance** $\rho(z_0, z_1)$ from $z_0$ to $z_1$ to be the infimum (greatest lower bound) of the hyperbolic lengths of all piecewise smooth curves in $\mathbb{D}$ from $z_0$ to $z_1$. Since conformal self-maps of $\mathbb{D}$ preserve the hyperbolic lengths of curves, they also preserve hyperbolic distances; that is, for any conformal self-map $w = f(z)$ of $\mathbb{D}$,

$$\rho(f(z_0), f(z_1)) = \rho(z_0, z_1), \qquad z_0, z_1 \in \mathbb{D}.$$

**Theorem.** *For any two distinct points $z_0, z_1$ in the open unit disk $\mathbb{D}$, there is a unique shortest curve in $\mathbb{D}$ from $z_0$ to $z_1$ in the hyperbolic metric, namely, the arc of the circle passing through $z_0$ and $z_1$ that is orthogonal to the unit circle.*

The paths of shortest hyperbolic length between points are called **hyperbolic geodesics**. The hyperbolic geodesics play the role that straight lines play in the Euclidean geometry of the plane. They satisfy all the axioms of Euclidean geometry except the parallel axiom (that through each point not on a given line there passes a unique straight line through the point and parallel to the given line).



hyperbolic geodesics

For a proof of the theorem, let $w = f(z)$ be a conformal self-map of $\mathbb{D}$ such that $f(z_0) = 0$. By multiplying by a unimodular constant, we can arrange that $f(z_1) = r > 0$. Since $f(z)$ preserves hyperbolic lengths, and since $f(z)$ maps circles orthogonal to the unit circle onto circles orthogonal to the unit circle, it suffices to show that the straight line segment from 0 to $r$ is a unique path of shortest hyperbolic length from 0 to $r$. For this, let $\gamma(t) = x(t) + iy(t)$, $0 \le t \le 1$, be a piecewise smooth path in $\mathbb{D}$ from 0 to $r$. Then $\alpha(t) = \operatorname{Re}(\gamma(t)) = x(t)$ defines a path in $\mathbb{D}$ from 0 to $r$ along the real axis, and

$$\int_\alpha \frac{|dz|}{1-|z|^2} = \int_0^1 \frac{|dx(t)|}{1-x(t)^2} \le \int_0^1 \frac{|dx(t)|}{1-|\gamma(t)|^2} \le \int_\gamma \frac{|dz|}{1-|z|^2}.$$

If $y(t) \neq 0$ for some $t$, then $|\gamma(t)| > |x(t)|$, and the first inequality above is strict. In this case, the path $\alpha(t)$ on the real axis is strictly shorter than the path $\gamma(t)$. Further, if $\alpha(t)$ is decreasing on some interval, we could reduce the integral by deleting a parameter interval over which $\alpha(t)$ starts and ends at the same value. We conclude that the integral is a minimum exactly when $\gamma(t)$ is real and nondecreasing, in which case the path is the straight line segment from 0 to $r$.

We turn now to an important reinterpretation of Pick's lemma.

**Theorem.** *Every analytic function $w = f(z)$ from the open unit disk $\mathbb{D}$ to itself is a contraction mapping with respect to the hyperbolic metric $\rho$,*

$$\rho(f(z_0), f(z_1)) \le \rho(z_0, z_1), \qquad z_0, z_1 \in \mathbb{D}. \tag{3.3}$$

*Further, there is strict inequality for all points $z_0, z_1 \in \mathbb{D}$, $z_0 \neq z_1$, unless $f(z)$ is a conformal self-map of $\mathbb{D}$, in which case there is equality for all $z_0, z_1 \in \mathbb{D}$.*

To see this, let $\gamma$ be the geodesic from $z_0$ to $z_1$. Then $f \circ \gamma$ is a curve from $f(z_0)$ to $f(z_1)$. Pick's lemma and the definition of the hyperbolic metric yield

$$\begin{aligned}\rho(f(z_0), f(z_1)) &\le 2\int_{f\circ\gamma} \frac{|dw|}{1-|w|^2} = 2\int_\gamma \frac{|f'(z)|\,|dz|}{1-|f(z)|^2} \\ &\le 2\int_\gamma \frac{|dz|}{1-|z|^2} = \rho(z_0, z_1).\end{aligned}$$

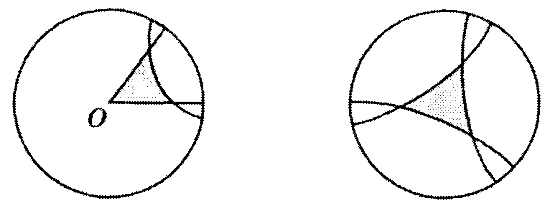
If $f(z)$ is not a conformal self-map of $\mathbb{D}$, there is strict inequality in Pick's lemma, and we obtain strict inequality in this estimate, hence in (3.3).

The hyperbolic distance from 0 to $z$ can be computed explicitly. It is

$$\rho(0, z) = 2\int_0^{|z|} \frac{dt}{1-t^2} = \int_0^{|z|} \left[\frac{1}{1-t} + \frac{1}{1+t}\right] dt = \log\left(\frac{1+|z|}{1-|z|}\right).$$

This shows that the hyperbolic distance from 0 to $z$ tends to $+\infty$ when $z$ tends to the boundary of the unit disk.

A **geodesic triangle** is an area bounded by three hyperbolic geodesics. Since the hyperbolic geodesics and the angles between them are preserved




hyperbolic triangles

by conformal self-maps of $\mathbb{D}$, we can map any geodesic triangle to a triangle with vertex at 0 and with the same angles between sides. For a geodesic triangle with vertex at 0, two of the sides are radial segments, and the third is an arc of a circle lying inside the Euclidean triangle with the two radii as sides. From this representation we see that the sum of the angles of any geodesic triangle is strictly less than $\pi$, which is the sum of the angles of the corresponding Euclidean triangle.

In connection with complex analysis, we have now been in contact with three spaces with strikingly different geometries. The first space is the complex plane $\mathbb{C}$ with the usual Euclidean metric $|dz|$. In the Euclidean plane, the geodesics are straight lines, and the sum of the angles of a geodesic triangle is exactly equal to $\pi$. The second space is the open unit disk $\mathbb{D}$ with the hyperbolic metric $2|dz|/(1-|z|^2)$. For the hyperbolic disk, the geodesics are arcs of circles orthogonal to the unit circle, and the sum of angles of a geodesic triangle is strictly less than $\pi$.

The third space is the extended complex plane $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$ with the spherical metric, which can be introduced in a manner completely analogous to the hyperbolic metric. Recall (Section I.3) that the chordal metric induced on $\mathbb{C}$ by the Euclidean metric of the sphere via the stereographic projection is given explicitly by

$$\text{chordal distance from } z \text{ to } w \;=\; \frac{2|z-w|}{\sqrt{1+|z|^2}\sqrt{1+|w|^2}}.$$

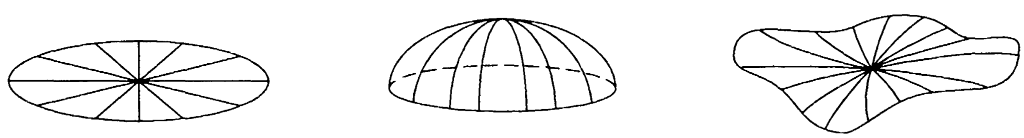
The infinitesimal form of this metric is $2|dz|/(1+|z|^2)$. If $\gamma$ is a path in $\mathbb{C}^*$, its **length in the spherical metric** is

$$\text{spherical length of } \gamma \;=\; 2\int_\gamma \frac{|dz|}{1+|z|^2} \;=\; 2\int \frac{|\gamma'(t)|}{1+|\gamma(t)|^2}\,dt\,.$$

This is the length of the corresponding path on the unit sphere in $\mathbb{R}^3$. The **distance from $z_1$ to $z_2$ in the spherical metric** is defined to be the infimum of the spherical lengths of the paths joining $z_1$ to $z_2$. Since the chordal metric is invariant under rotations of the sphere, so is the spherical metric, and consequently, the lengths of paths and the distances between points in the spherical metric are invariant under rotations. It is not difficult to show that the geodesics in the spherical metric correspond to great circles on the sphere, and the sum of the angles of a geodesic triangle is strictly greater than $\pi$.

Each of these three spaces is homogeneous, in the sense that any prescribed point can be transported to any other by an "isometry." Thus for each of these spaces, any scalar quantity that is invariant under isometries is constant. It turns out that a notion of "scalar curvature" can be associated to each of the spaces (see the exercises), and the curvature is invariant under isometries, so that in each case the curvature is constant. The Euclidean plane has constant zero curvature, the sphere has constant positive curvature, and the hyperbolic disk has constant negative curvature. The curvature can be related to the area and the sum of angles of geodesic triangles (Gauss-Bonnet formula). We summarize these properties in tabular form.

| Geometry | Euclidean | Spherical | Hyperbolic |
|---|---|---|---|
| Infinitesimal length | $\lvert dz\rvert$ | $\dfrac{2\lvert dz\rvert}{1+\lvert z\rvert^2}$ | $\dfrac{2\lvert dz\rvert}{1-\lvert z\rvert^2}$ |
| Oriented isometries | $e^{i\varphi}z+b$ | rotations | conformal self-maps |
| Curvature | $0$ | $+1$ | $-1$ |
| Geodesics | lines | great circles | circles $\perp$ unit circle |
| Angles of triangle | $=\pi$ | $>\pi$ | $<\pi$ |
| Disk circumference | $2\pi\rho$ | $2\pi\rho-\dfrac{\pi\rho^3}{3}+\mathcal{O}(\rho^5)$ | $2\pi\rho+\dfrac{\pi\rho^3}{3}+\mathcal{O}(\rho^5)$ |



Euclidean disk spherical disk hyperbolic disk

## Exercises for IX.3

1. Show by direct computation that $|w'(z)| = (1-|w|^2)/(1-|z|^2)$ for any conformal self-map $w = f(z)$ of $\mathbb{D}$.

2. A **hyperbolic disk** centered at $z_0 \in \mathbb{D}$ of radius $\rho > 0$ consists of all $z \in \mathbb{D}$ such that $\rho(z, z_0) < \rho$. (a) Show that the hyperbolic disk centered at 0 of radius $\rho$ is a Euclidean disk of radius $r = (e^\rho - 1)/(e^\rho + 1)$. (b) Show that any hyperbolic disk is a Euclidean disk.

3. Denote by $c(z,\rho)$ and $r(z,\rho)$ the Euclidean center and Euclidean radius of the hyperbolic disk centered at $z$ of hyperbolic radius $\rho$. (a) For fixed $\rho$, show that $r(z,\rho)/(1-|z|)$ tends to a constant $A > 0$ as $|z| \to 1$. (b) For fixed $\rho$, show that $|z - c(z,\rho)|/r(z,\rho)$ tends to a constant $B$, $0 < B < 1$, as $|z| \to 1$.

4. Show that the circumference of a hyperbolic disk of radius $\rho$ is $2\pi \sinh \rho$. *Hint.* Show first that the hyperbolic circumference of a Euclidean disk of radius $r$ centered at 0 is $4\pi r/(1-r^2)$.

5. We define the **hyperbolic area** of a subset $E$ of $\mathbb{D}$ to be

$$4 \iint_E \frac{dx\,dy}{(1-|z|^2)^2}.$$

Show that the hyperbolic area is invariant under conformal self-maps of $\mathbb{D}$. Show that the hyperbolic area of a hyperbolic disk of radius $\rho$ is given by

$$2\pi(\cosh \rho \; - \; 1) \;=\; \pi\rho^2 + \frac{\pi}{12}\rho^4 + \mathcal{O}(\rho^6).$$

6. Establish the following, for the spherical metric. (a) The circumference of a spherical disk of radius $\rho$ is $2\pi \sin \rho$, $0 < \rho < \pi$. (b) The area of a spherical disk of radius $\rho$ is given by

$$2\pi(1 - \cos \rho) \;=\; \pi\rho^2 - \frac{\pi}{12}\rho^4 + \mathcal{O}(\rho^6).$$

(c) The geodesics in the spherical metric correspond to great circles on the sphere. *Hint.* It suffices to show that the shortest curve from 0 to $\varepsilon$ in the spherical metric is the straight line segment joining them.

7. Show that an isometry of the hyperbolic disk $\mathbb{D}$ is either a conformal self-map of $\mathbb{D}$ or the composition of a conformal self-map and the reflection $z \mapsto \bar{z}$.

8. Let $f(z) = (az+b)/(cz+d)$, where $ad - bc = 1$. Show that $f(z)$ is an isometry in the spherical metric if and only if the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is unitary.

9. Show that the function $f(z) = z^2$ is strictly contracting with respect to the hyperbolic metric on any subdisk $\{|z| \le r\}$, $0 < r < 1$, and that any branch of the square root function is strictly expanding, by establishing the following. (a) For fixed $r$, $0 < r < 1$, show that

$$\rho(z^2, \zeta^2) \;\le\; \frac{2r}{1+r^2}\rho(z,\zeta), \qquad |z|, |\zeta| \le r.$$

When does equality hold? (b) Show that the constant $2r/(1+r^2)$ in (a) is sharp. (c) For fixed $s$, $0 < s < 1$, show that

$$\rho\left(\pm\sqrt{z}, \pm\sqrt{\zeta}\right) \;\ge\; \frac{1+s}{2\sqrt{s}}\rho(z,\zeta), \qquad |z|, |\zeta| \le s.$$

10. Show that

$$d(z,w) \;=\; \left|\frac{z-w}{1-\bar{w}z}\right|, \qquad |z|,|w|<1,$$

satisfies the triangle inequality, that is, $d(z,w) \le d(z,\zeta)+d(\zeta,w)$ for all $z,\zeta,w \in \mathbb{D}$. *Remark.* This can be regarded as the analogue of the chordal metric for the sphere (defined in Section I.3). Except for the constant factor 2, the hyperbolic metric is the infinitesimal version of the metric function $d(z,w)$.

11. Show that the metric function $d(z,w)$ defined in the preceding exercise satisfies

$$d(f(z),f(w)) \le d(z,w), \qquad |z|,|w|<1,$$

for any analytic function $f(z)$ from $\mathbb{D}$ to $\mathbb{D}$. Show that equality obtains whenever $f(z)$ is a conformal self-map of $\mathbb{D}$, and otherwise there is strict inequality for all $z \ne w$.

12. A conformal map $g(z)$ of a domain $D$ onto the open unit disk $\mathbb{D}$ induces the metric $\rho_D$ on $D$ defined by

$$d\rho_D(z) = \frac{2|g'(z)|}{1-|g(z)|^2}|dz|, \qquad z \in D.$$

Show that $\rho_D$ is independent of the conformal map $g(z)$ of $D$ onto $\mathbb{D}$. *Remark.* The metric $\rho_D$ is called the **hyperbolic metric** of the simply connected domain $D$.

13. Show that the hyperbolic metric of the upper half-plane $\mathbb{H}$ is given by

$$d\rho_{\mathbb{H}}(z) = \frac{|dz|}{y}, \qquad z = x+iy,\; y>0.$$

What are the geodesics in the hyperbolic metric? Illustrate with a sketch.

14. Show that the horizontal strip $S = \{-\pi/2 < \operatorname{Im} z < \pi/2\}$ has hyperbolic metric

$$d\rho_S(z) = \frac{|dz|}{\cos y}, \qquad z = x+iy,\; -\pi/2 < y < \pi/2.$$

Sketch the hyperbolic geodesics that are orthogonal to the vertical interval $\{iy : -\pi/2 < y < \pi/2\}$.

15. The **curvature** of the metric $\sigma(z)|dz|$ is defined to be

$$\kappa(z) \;=\; -\frac{1}{\sigma(z)^2}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\log\sigma(z).$$

Find the curvature of each of the spherical, the hyperbolic, and the Euclidean metrics.

16. (Wolff-Denjoy Theorem.) Let $f(z)$ be an analytic function from $\mathbb{D}$ to $\mathbb{D}$. Let $f_n(z)$ denote the $n$th iterate of $f(z)$, and let $K_r$ denote the closed disk $\{|z| \leq r\}$.
    (a) Show that if $f(z)$ is not a conformal self-map of $\mathbb{D}$, then for any $r < 1$ there is a constant $c < 1$ such that $\rho(f(z), f(w)) \leq c\rho(z, w)$ for $z, w \in K_r$.
    (b) Show that if the image $f(\mathbb{D})$ is contained in $K_r$ for some $r < 1$, then the iterates $f_n(z)$ converge uniformly on $\mathbb{D}$ to a fixed point for $f(z)$.
    (c) Show that if $f(z)$ is not a conformal self-map of $\mathbb{D}$, and if there is $r < 1$ such that the iterates of some point $z_0 \in \mathbb{D}$ visit $K_r$ infinitely often, then the iterates $f_n(z)$ converge normally on $\mathbb{D}$ to a fixed point of $f(z)$. *Hint.* First find the fixed point.
    (d) Show that if the iterates of some point $z_0 \in \mathbb{D}$ tend to the unit circle $\partial\mathbb{D}$, then there is a point $\zeta \in \partial\mathbb{D}$ (the **Wolff-Denjoy point**) such that the iterates $f_n(z)$ converge normally on $\mathbb{D}$ to $\zeta$. *Hint.* Suppose $z_0 = 0$. Define $g_\varepsilon(z) = (1 - \varepsilon)f(z)$, let $z_\varepsilon$ be the fixed point of $g_\varepsilon(z)$, and let $D_\varepsilon$ be the hyperbolic disk centered at $z_\varepsilon$ with 0 on its boundary. Show that the limit $D$ of the $D_\varepsilon$'s is a Euclidean disk that is invariant under $f(z)$ and whose boundary meets $\partial\mathbb{D}$ in exactly one point.

# X

# Harmonic Functions and the Reflection Principle

In Section 1 we introduce the Poisson kernel function and we develop the Poisson integral representation for harmonic functions on the open unit disk. The Poisson kernel is the analogue for harmonic functions of the Cauchy kernel for analytic functions, and the Poisson integral formula solves the Dirichlet problem for the unit disk. In Section 2 we use this solution to characterize harmonic functions by the mean value property. This characterization is the analogue of Morera's theorem characterizing analytic functions. In Section 3 we apply the characterization of harmonic functions to establish the Schwarz reflection principle for harmonic functions. The reflection principle plays a key role in the study of boundary behavior of conformal maps.

## 1. The Poisson Integral Formula

We wish to extend a given continuous complex-valued function $h\left(e^{i\theta}\right)$ on the unit circle continuously to the closed unit disk $\{|z| \leq 1\}$ so as to be harmonic on the interior of the disk. Any such extension is unique, since the difference of two such extensions is zero on the boundary, hence zero on the entire disk, by the maximum principle. Our strategy is to derive a formula for the extension in the case that $h\left(e^{i\theta}\right)$ is a trigonometric polynomial, and then to show that the formula provides an extension even when $h\left(e^{i\theta}\right)$ is only continuous.

We start with the trigonometric monomial $e^{ik\theta}$. A harmonic extension is given explicitly by $r^{|k|}e^{ik\theta}$. If $k \geq 0$ this extension is $r^k e^{ik\theta} = z^k$, which is analytic. If $k < 0$ this extension is $r^{-k}e^{ik\theta} = \bar{z}^{(-k)}$, which is conjugate-analytic hence harmonic. Proceeding by linearity, we see that the trigonometric polynomial $h\left(e^{i\theta}\right) = \sum_{k=-N}^{N} a_k e^{ik\theta}$ has the (unique) harmonic extension

$$\tilde{h}\left(re^{i\theta}\right) \;=\; \sum_{k=-N}^{N} a_k r^{|k|} e^{ik\theta}.$$

We capture the coefficient $a_m$ by multiplying $h\left(e^{i\theta}\right)$ by $e^{-im\theta}$ and integrating. The orthogonality relations for complex exponentials (Section VI.6) yield

$$a_m \;=\; \int_{-\pi}^{\pi} h\left(e^{i\theta}\right) e^{-im\theta} \frac{d\theta}{2\pi}.$$

Substituting this expression into the formula for $\tilde{h}\left(re^{i\theta}\right)$, we obtain

$$\begin{aligned}\tilde{h}(re^{i\theta}) &= \sum_{k=-\infty}^{\infty} \left(\int_{-\pi}^{\pi} h\left(e^{i\varphi}\right) e^{-ik\varphi} \frac{d\varphi}{2\pi}\right) r^{|k|} e^{ik\theta} \\ &= \int_{-\pi}^{\pi} h\left(e^{i\varphi}\right) \left[\sum_{k=-\infty}^{\infty} r^{|k|} e^{-ik\varphi} e^{ik\theta}\right] \frac{d\varphi}{2\pi}.\end{aligned}$$

In order to simplify this expression, we introduce the **Poisson kernel function** defined by

$$P_r(\theta) \;=\; \sum_{k=-\infty}^{\infty} r^{|k|} e^{ik\theta}. \tag{1.1}$$

For each fixed $\rho < 1$, this series converges uniformly for $r \leq \rho$ and $-\pi \leq \theta \leq \pi$, by the Weierstrass $M$-test, since then $\left|r^{|k|}e^{ik\theta}\right| \leq \rho^{|k|}$. In terms of the Poisson kernel, the formula for the harmonic extension $\tilde{h}\left(re^{i\theta}\right)$ becomes

$$\tilde{h}(re^{i\theta}) \;=\; \int_{-\pi}^{\pi} h\left(e^{i\varphi}\right) P_r(\theta-\varphi) \frac{d\varphi}{2\pi}, \qquad re^{i\theta} \in \mathbb{D}. \tag{1.2a}$$

If we make a change of variable $\varphi \mapsto \theta - \varphi$ and use the $2\pi$-periodicity of $P_r(\theta)$, we obtain an alternative form of (1.2a),

$$\tilde{h}(re^{i\theta}) \;=\; \int_{-\pi}^{\pi} h\left(e^{i(\theta-\varphi)}\right) P_r(\varphi) \frac{d\varphi}{2\pi}, \qquad re^{i\theta} \in \mathbb{D}. \tag{1.2b}$$

Now we look at the Poisson kernel function $P_r(\theta)$ more closely. Setting $z = re^{i\theta}$ and setting $j = -k$ when $k < 0$ in (1.1), we obtain

$$P_r(\theta) \;=\; 1 + \sum_{k=1}^{\infty} z^k + \sum_{j=1}^{\infty} \bar{z}^j.$$

Summing these two geometric series, we obtain

$$P_r(\theta) \;=\; 1 + \frac{z}{1-z} + \frac{\bar{z}}{1-\bar{z}}, \qquad z = re^{i\theta} \in \mathbb{D}. \tag{1.3}$$

Putting this over the common denominator

$$|1-z|^2 \;=\; (1-z)(1-\bar{z}) \;=\; 1+r^2-2r\cos\theta,$$

we obtain

$$(1.4) \qquad P_r(\theta) \;=\; \frac{1-|z|^2}{|1-z|^2} \;=\; \frac{1-r^2}{1+r^2-2r\cos\theta}, \qquad z = re^{i\theta} \in \mathbb{D}.$$

Since the Poisson kernel is $2\pi$-periodic, we focus on its behavior on the period interval $-\pi \le \theta \le \pi$. From the Poisson integral representation formula (1.2b) for the constant function $h \equiv 1$, we have

$$(1.5) \qquad \int_{-\pi}^{\pi} P_r(\theta)\frac{d\theta}{2\pi} \;=\; 1.$$

Three other properties of the Poisson kernel, which follow immediately from the formula (1.4), are

$$(1.6) \qquad P_r(\theta) > 0, \qquad -\pi \le \theta \le \pi,$$

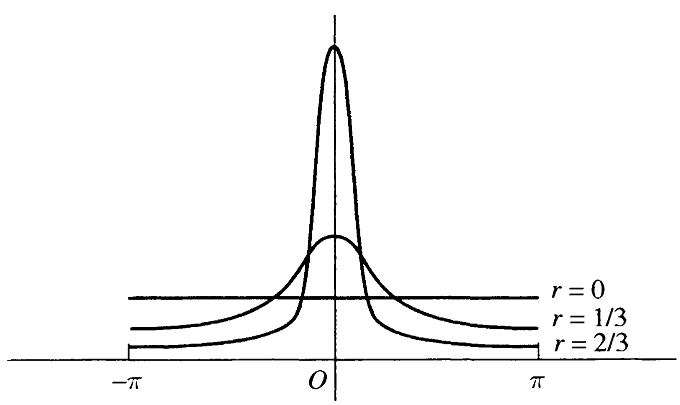
$$(1.7) \qquad P_r(-\theta) = P_r(\theta), \qquad -\pi \le \theta \le \pi,$$

(1.8) $P_r(\theta)$ is increasing for $-\pi \le \theta \le 0$ and decreasing for $0 \le \theta \le \pi$.

Since $P_r(\delta) \to 0$ as $r \to 1$ for each fixed $\delta > 0$, we obtain from (1.8) the following key property of the Poisson kernel:

(1.9) for fixed $\delta > 0$, $\max\{\, P_r(\theta) \,:\, \delta \le |\theta| \le \pi \,\} \to 0$ as $r \to 1$.

These five properties, (1.5) through (1.9), are fundamental for understanding the Poisson kernel. Properties (1.5) and (1.6) together show that each $P_r(\theta)d\theta/2\pi$ is a "probability measure," that is, a positive "mass distribution" with unit total mass. Properties (1.5), (1.6), and (1.9) show that the family of functions $P_r(\theta)$ form an "approximate identity," in the sense that the mass of the probability measure $P_r(\theta)d\theta/2\pi$ concentrates at the point $\theta = 0$ as $r \to 1$. Identity (1.7) reflects the symmetry of the Poisson kernel.




Another important property of the Poisson kernel function is that $P_r(\theta)$ is a harmonic function of $z = re^{i\theta}$ for $z \in \mathbb{D}$. In fact, (1.3) yields an explicit representation of the Poisson kernel function as the real part of an analytic function,

$$P_r(\theta) \;=\; 1 + 2\,\mathrm{Re}\left(\frac{z}{1-z}\right) \;=\; \mathrm{Re}\left(\frac{1+z}{1-z}\right), \qquad z = re^{i\theta} \in \mathbb{D}.$$

The Poisson integral formula (1.2a) becomes

$$\tilde{h}(re^{i\theta}) \;=\; \int_{-\pi}^{\pi} h\left(e^{i\varphi}\right) \mathrm{Re}\left(\frac{1+re^{i(\theta-\varphi)}}{1-re^{i(\theta-\varphi)}}\right) \frac{d\varphi}{2\pi}.$$

Thus

$$\tilde{h}(z) \;=\; \int_{-\pi}^{\pi} h\left(e^{i\varphi}\right) \mathrm{Re}\left(\frac{e^{i\varphi}+z}{e^{i\varphi}-z}\right) \frac{d\varphi}{2\pi}, \qquad z \in \mathbb{D}. \tag{1.10}$$

If we substitute for $h\left(e^{i\theta}\right)$ a real-valued trigonometric polynomial $u\left(e^{i\theta}\right)$ in this formula, we may take real parts after integrating, and we obtain

$$\tilde{u}(z) \;=\; \mathrm{Re}\int_{-\pi}^{\pi} u(e^{i\varphi})\, \frac{e^{i\varphi}+z}{e^{i\varphi}-z}\, \frac{d\varphi}{2\pi}, \qquad z \in \mathbb{D}. \tag{1.11}$$

The integral depends analytically on the parameter $z$. Thus (1.11) expresses the Poisson integral $\tilde{u}(z)$ as the real part of an explicit analytic function.

Now we change our point of view. Instead of focusing on trigonometric polynomials, we consider an arbitrary continuous complex-valued function $h\left(e^{i\theta}\right)$ on the unit circle. We define the **Poisson integral** $\tilde{h}(z)$ of $h\left(e^{i\theta}\right)$ to be the function on the open unit disk $\mathbb{D}$ given by

$$\tilde{h}(z) \;=\; \int_{-\pi}^{\pi} h\left(e^{i\varphi}\right) P_r(\theta-\varphi)\frac{d\varphi}{2\pi}, \qquad z = re^{i\theta} \in \mathbb{D}. \tag{1.12}$$

This is just (1.2a), and it can be rewritten as (1.2b). Formula (1.10) still holds, and for real-valued continuous functions $u\left(e^{i\theta}\right)$, formula (1.11) holds.

The correspondence $h \mapsto \tilde{h}$ is linear, that is, the Poisson integral of $c_1h_1 + c_2h_2$ is $c_1\tilde{h}_1 + c_2\tilde{h}_2$. Also, observe that the maximum principle holds, in the sense that if $|h\left(e^{i\theta}\right)| \le M$ for all $\theta$, then $|\tilde{h}(z)| \le M$ for all $z \in \mathbb{D}$. This follows from (1.5) and (1.6), that is, from the fact that the Poisson kernel determines a probability measure.

**Theorem.** *Let $h\left(e^{i\theta}\right)$ be a continuous function on the unit circle. Then the Poisson integral $\tilde{h}(z)$ defined by* (1.12) *is a harmonic function on the open unit disk that has boundary values $h\left(e^{i\theta}\right)$, that is, $\tilde{h}(z)$ tends to $h(\zeta)$ as $z \in \mathbb{D}$ tends to $\zeta \in \partial\mathbb{D}$.*

The harmonicity can be checked by differentiating under the integral sign in (1.12). It can also be seen by decomposing $h\left(e^{i\theta}\right)$ into its real and imaginary parts, $h\left(e^{i\theta}\right) = u\left(e^{i\theta}\right) + iv\left(e^{i\theta}\right)$. Formula (1.11) shows that

$\tilde{u}(z)$ and $\tilde{v}(z)$ are harmonic, and hence $\tilde{h}(z) = \tilde{u}(z) + i\tilde{v}(z)$ is harmonic for $|z| < 1$.

Since the statement about the boundary values is important, we sketch two proofs. The first proof depends upon the fact that any continuous function on the unit circle can be approximated uniformly by trigonometric polynomials. (One way to see this is to approximate the continuous function by a smooth function and then to appeal to the result proved in Section VI.6 that the Fourier series of any smooth function on the unit circle converges uniformly to the function.) The second proof illustrates how the properties of an approximate identity are typically used.

The first proof boils down to the fact that a uniform limit of continuous functions is continuous. It runs as follows. Let $h\left(e^{i\theta}\right)$ be continuous, and let $\varepsilon > 0$. Let $g\left(e^{i\theta}\right) = \sum a_k e^{ik\theta}$ be a trigonometric polynomial such that $\left|h\left(e^{i\theta}\right) - g\left(e^{i\theta}\right)\right| \leq \varepsilon$ for all $e^{i\theta} \in \partial\mathbb{D}$. Then $\left|\tilde{h}(z) - \tilde{g}(z)\right| \leq \varepsilon$ for all $z \in \mathbb{D}$. Now, $\tilde{g}\left(re^{i\theta}\right) = \sum a_k r^{|k|} e^{ik\theta}$ attains the boundary values $g\left(e^{i\theta}\right)$ continuously on $\partial\mathbb{D}$. Hence the composite function equal to $h\left(e^{i\theta}\right)$ on $\partial\mathbb{D}$ and $\tilde{h}\left(e^{i\theta}\right)$ on $\mathbb{D}$ can be approximated to within $\varepsilon$ by a continuous function. Consequently, the values $\tilde{h}(z)$ cluster to within $\varepsilon$ of $h\left(e^{i\theta}\right)$ as $z \in \mathbb{D}$ tends to $e^{i\theta} \in \partial\mathbb{D}$. Since $\varepsilon > 0$ is arbitrary, $\tilde{h}(z)$ tends to $h\left(e^{i\theta}\right)$ as $z \in \mathbb{D}$ tends to $e^{i\theta} \in \partial\mathbb{D}$.

The second proof depends only on properties (1.5), (1.6), and (1.9) of the Poisson kernel. Again let $\varepsilon > 0$. Choose $M > 0$ such that $\left|h\left(e^{i\theta}\right)\right| \leq M$. Since $h\left(e^{i\theta}\right)$ is uniformly continuous, we can choose $\delta > 0$ so small that $\left|h\left(e^{i\theta}\right) - h\left(e^{i\varphi}\right)\right| < \varepsilon$ whenever $|\theta - \varphi| < \delta$. Using (1.5) we have

$$\tilde{h}(z) - h\left(e^{i\theta}\right) = \int_{-\pi}^{\pi} \left[h\left(e^{i(\theta-\varphi)}\right) - h\left(e^{i\theta}\right)\right] P_r(\varphi)\frac{d\varphi}{2\pi}.$$

Now we take absolute values, break the integral cleverly into pieces, and make the obvious estimate on each piece, to obtain

$$\begin{aligned} \left|\tilde{h}\left(re^{i\theta}\right) - h\left(e^{i\theta}\right)\right| &\leq \left(\int_{-\delta}^{\delta} + \int_{\delta\leq|\varphi|\leq\pi}\right)\left|h\left(e^{i(\theta-\varphi)}\right) - h\left(e^{i\theta}\right)\right| P_r(\varphi)\frac{d\varphi}{2\pi} \\ &\leq \int_{-\delta}^{\delta} \varepsilon P_r(\varphi)\frac{d\varphi}{2\pi} + 2M \max_{\delta\leq|\varphi|\leq\pi} P_r(\varphi) \int_{\delta\leq|\varphi|\leq\pi} \frac{d\varphi}{2\pi} \\ &\leq \varepsilon + 2M \max_{\delta\leq|\varphi|\leq\pi} P_r(\varphi). \end{aligned}$$

By (1.9), the summand on the right tends to 0 as $r \to 1$. Thus again the values $\tilde{h}(z)$ cluster to within $\varepsilon$ of $h\left(e^{i\theta}\right)$ as $z \to e^{i\theta}$, this for any $\varepsilon > 0$, so $\tilde{h}(z) \to h\left(e^{i\theta}\right)$ as $z \to e^{i\theta}$.

## Exercises for X.1

1. Show that the Fourier coefficients of $P_r(\theta)$ are $c_k = r^{|k|}$, $-\infty < k < \infty$. (Here we fix $r$, $0 \leq r < 1$, and we regard $P_r(\theta)$ as a function on the circle.)

2. Let $R > 0$, and let $h\left(Re^{i\theta}\right)$ be a continuous function on the circle $\{|z| = R\}$. Show that the function
$$\tilde{h}(z) \;=\; \int_{-\pi}^{\pi} \frac{R^2 - r^2}{R^2 + r^2 - 2rR\cos(\theta - \varphi)} h\left(Re^{i\varphi}\right) \frac{d\varphi}{2\pi}, \qquad |z| < R,$$
is harmonic on the disk $\{|z| < R\}$ and has boundary values $h\left(Re^{i\theta}\right)$ on the boundary circle.

3. Suppose that $f(z) = u(z) + iv(z)$ is analytic for $|z| < 1$ and that $u(z)$ extends to be continuous on the closed disk $\{|z| \leq 1\}$. Show that
$$f(z) \;=\; \int_0^{2\pi} u(e^{i\varphi}) \frac{e^{i\varphi} + z}{e^{i\varphi} - z} \frac{d\varphi}{2\pi} + iv(0), \qquad |z| < 1.$$
*Remark.* This is the **Schwarz formula**, expressing an analytic function in terms of the boundary values of its real part.

4. Let $\{f_n(z) = u_n(z) + iv_n(z)\}$ be a sequence of analytic functions on the open unit disk $\mathbb{D}$ such that $u_n(z)$ extends continuously to $\partial\mathbb{D}$, $u_n\left(e^{i\theta}\right)$ converges uniformly on $\partial\mathbb{D}$ to $u\left(e^{i\theta}\right)$, and $v_n(0)$ converges. Show that $f_n(z)$ converges normally on $\mathbb{D}$ to an analytic function $f(z)$ whose real part is $\tilde{u}(z)$. *Hint.* Use the Schwarz formula.

5. Let $h\left(e^{i\theta}\right)$ be a piecewise continuous function (or an integrable function) on the unit circle. Show that the Poisson integral $\tilde{h}(z)$ tends to $h(\zeta)$ as $z \in \mathbb{D}$ tends to any point $\zeta$ of the unit circle at which $h\left(e^{i\theta}\right)$ is continuous.

6. A function $f(z)$, $z \in \mathbb{D}$, is said to have **radial limit** $L$ at $\zeta \in \partial\mathbb{D}$ if $f(r\zeta) \to L$ as $r$ increases to 1. Let $h\left(e^{i\theta}\right)$ be a piecewise continuous function on the unit circle. Show that $\tilde{h}(z)$ has a radial limit at each $\zeta \in \partial\mathbb{D}$, equal to the average of the limits of $h\left(e^{i\theta}\right)$ at $\zeta$ from each side.

7. For each $t > 0$, define the kernel function $C_t(s)$ on the real line by
$$C_t(s) \;=\; \frac{t}{\pi} \frac{1}{s^2 + t^2}, \qquad -\infty < s < \infty.$$
For $h(\xi)$ a bounded piecewise continuous function on the real line, define a function $\tilde{h}(s + it)$ on the open upper half-plane $\mathbb{H}$ by
$$\tilde{h}(s + it) \;=\; \int_{-\infty}^{\infty} C_t(s - \xi) h(\xi)\, d\xi, \qquad s + it \in \mathbb{H}.$$

(a) Sketch the graph of $C_t(s)$ for $t = 1$, $t = 0.1$, and $t = 0.01$. (b) Show that $C_t(s) > 0$ and $\int_{-\infty}^{\infty} C_t(s)\,ds = 1$. (Thus $C_t(s)ds$ is a probability measure.) (c) Show that for each $\delta > 0$, $\int_{\{|s|>\delta\}} C_t(s)\,ds \to 0$ as $t \to 0$. (Thus $C_t(s)ds$ is an approximate identity.) (d) Show that $\tilde{h}(s+it)$ is a bounded harmonic function in the upper half-plane. (e) Show that $\tilde{h}(s+it) \to h(\xi_0)$ as $s+it \in \mathbb{H}$ tends to $\xi_0$, whenever $h(\xi)$ is continuous at $\xi_0$. *Remark.* The kernel function $C_t(s)$ is the **Poisson kernel** for the upper half-plane.

8. Let $h\left(e^{i\varphi}\right)$ be a continuous complex-valued function on $\partial\mathbb{D}$, and let $\tilde{h}(z)$ be its Poisson integral.
   (a) Show that

$$\frac{\partial^m}{\partial z^m}\tilde{h}(z) = m!\int_{-\pi}^{\pi} h(e^{i\varphi})\frac{e^{i\varphi}}{(e^{i\varphi}-z)^{m+1}}\frac{d\varphi}{2\pi}, \qquad z \in \mathbb{D},$$
$$\frac{\partial^m}{\partial \bar{z}^m}\tilde{h}(z) = m!\int_{-\pi}^{\pi} h(e^{i\varphi})\frac{e^{-i\varphi}}{(e^{-i\varphi}-\bar{z})^{m+1}}\frac{d\varphi}{2\pi}, \qquad z \in \mathbb{D}.$$

   *Hint.* Use the identity

$$P_r(\theta-\varphi) = 1 + \frac{ze^{-i\varphi}}{1-ze^{-i\varphi}} + \frac{\bar{z}e^{i\varphi}}{1-\bar{z}e^{i\varphi}}, \qquad z = e^{i\theta},$$

   and justify differentiating under the integral sign.
   (b) Show that if $\left|h\left(e^{i\theta}\right)\right| \le M$ on $\partial\mathbb{D}$, then

$$\left|\frac{\partial^m}{\partial z^m}\tilde{h}(z)\right| \le m!M\int_{-\pi}^{\pi}\frac{1}{|e^{i\varphi}-\rho|^{m+1}}\frac{d\varphi}{2\pi}, \qquad |z| \le \rho,$$

   with similar estimates for the $\bar{z}$-derivatives.
   (c) Show that if $h\left(e^{i\theta}\right) = e^{-i\theta}\left(e^{i\theta}-z\right)^{m+1}/|e^{i\theta}-z|^{m+1}$, then equality holds in (b) when $|z| = \rho$.

9. Let $\{h_n(e^{i\varphi})\}$ be a sequence of continuous functions on the unit circle $\partial\mathbb{D}$ that converges uniformly to $h(e^{i\varphi})$. Show that for each fixed $\rho < 1$, the partial derivatives of $\tilde{h}_n(z)$ converge to the corresponding partial derivatives of $\tilde{h}(z)$ uniformly for $|z| \le \rho$. *Remark.* It suffices to show this for the partial derivatives $\partial^m/\partial z^m$ and $\partial^m/\partial\bar{z}^m$. See Exercise IV.8.4.

## 2. Characterization of Harmonic Functions

Recall (Section III.4) that a continuous function $h(z)$ on a domain $D$ has the mean value property if for each $z_0 \in D$, the value $h(z_0)$ is the average of $h(z)$ over any small circle centered at $z_0$. We have seen that harmonic functions have the mean value property. Here we prove the converse.

**Theorem.** *Let $h(z)$ be a continuous function on a domain $D$. Then $h(z)$ is harmonic on $D$ if and only if $h(z)$ has the mean value property on $D$.*

The proof depends on solving the Dirichlet problem for disks in $D$. Roughly speaking, the **Dirichlet problem** for a domain $U$ with boundary $\partial U$ is to extend a given function $f(\zeta)$ on $\partial U$ to a harmonic function $\tilde{f}(z)$ on $U$, so that the harmonic extension $\tilde{f}(z)$ has boundary values $f(\zeta)$. Typically we assume that $f(\zeta)$ is a continuous function on $\partial U$, and we ask that $\tilde{f}(z)$ attain the boundary values $f(\zeta)$ continuously, in the sense that $\tilde{f}(z)$ tends to $f(\zeta)$ as $z \in U$ tends to $\zeta \in \partial U$. In the preceding section we showed that the Poisson integral solves the Dirichlet problem for the open unit disk $\mathbb{D}$. By making a change of variable $z \mapsto az + b$, we can solve the Dirichlet problem on any disk. An explicit solution is obtained by changing variable in the Poisson integral formula. (See Exercise 1.2.)

So let $U$ be any open disk in $D$ whose boundary is also contained in $D$. Since the Dirichlet problem is solvable for $U$, there is a harmonic function $g(z)$ on $U$ that has boundary values $h(z)$ on $\partial U$. Then $h(z) - g(z)$ is a continuous function on $U$ that has the mean value property on $U$, and $h(z) - g(z)$ tends to 0 on $\partial U$. By the maximum principle for functions with the mean value property (Section III.4), $h(z) - g(z) = 0$ on $U$. Consequently, $h(z) = g(z)$ on $U$, and $h(z)$ is harmonic on $U$. Since this holds for all $U$, $h(z)$ is harmonic on $D$.

The characterization of harmonic functions by the mean value property is the analogue for harmonic functions of Morera's theorem for analytic functions. The hypothesis involves an integration, which is easier to perform than differentiation. The characterization can be exploited in the same way as the characterization of analytic functions by Morera's theorem. For instance, it can be used to give a simple proof that the limit of a uniformly convergent sequence of harmonic functions is harmonic, since it is easy to check that the uniform limit of functions with the mean value property has the mean value property. It can also be used (Exercise 1) to show that an integral of a function that depends harmonically on a parameter also depends harmonically on the parameter.

It is rather striking that a continuous function with the mean value property automatically has partial derivatives of all orders.

## Exercises for X.2

1. Let $g(t, z)$ be a continuous function defined for $a \leq t \leq b$ and $z$ in a domain $D$, and suppose that $g(t, z)$ is a harmonic function of $z$ for each fixed $t$. Show that

$$G(z) = \int_a^b g(t, z)dt, \qquad z \in D,$$

is harmonic on $D$.

2. Assume that $u(x,y)$ is a twice continuously differentiable function on a domain $D$. Let $(x_0,y_0) \in D$, and let $A_\varepsilon(x_0,y_0)$ be the average of $u(x,y)$ on the circle centered at $(x_0,y_0)$ of radius $\varepsilon$. Show that
$$\lim_{\varepsilon\to 0} \frac{A_\varepsilon(x_0,y_0) - u(x_0,y_0)}{\varepsilon^2} = \frac{1}{4}\Delta u(x_0,y_0),$$
where $\Delta$ is the Laplacian operator (Section II.5).

3. For fixed $\rho > 0$, define $h(z) = e^{i\rho \operatorname{Im} z}$. Show that if $\rho$ is a zero of the Bessel function $J_0(z)$, then the value of $h(z)$ at any point is its average over the circle of radius $r = 1$ centered at the point,
$$h(z) = \int_0^{2\pi} h\left(z + e^{i\theta}\right)\frac{d\theta}{2\pi}, \qquad z \in \mathbb{C}.$$
Use the Schlömilch formula (Exercise VI.1.3).

4. Suppose that $r_1, r_2 > 0$ are such that $r_2/r_1$ is a quotient of two positive zeros of the Bessel function $J_0(z)$. Show that there is a continuous function on the complex plane such that the averages of the function over the circles of radius $r_1$ and $r_2$ centered at any point $z$ coincide with the value of the function at $z$, yet the function is not harmonic. Use the fact (easily derived from the differential equation in Section V.4) that $J_0(z)$ has zeros on the positive real axis. *Remark.* The condition is sharp. If $r_2/r_1$ is not the quotient of two positive zeros of the Bessel function $J_0(z)$, then any continuous function on the complex plane that has the mean value property for circles of radius $r_1$ and $r_2$ is harmonic, by a theorem of L. Zalcman.

5. Suppose that $r_1, r_2 > 0$ are such that $r_2/r_1$ is a zero of the Bessel function $J_1(z)$. Show that there is a continuous function $f(z)$ on the complex plane such that $\int_\gamma f(z)\,dz = 0$ for all circles in the complex plane of radius $r_1$ and $r_2$, yet $f(z)$ is not analytic. *Suggestion.* Proceed as in Exercises 3 and 4, and use the fact that $J_1(z)$ has zeros on the positive real axis. Again it turns out that the condition is sharp.

## 3. The Schwarz Reflection Principle

Suppose a given function is analytic on a domain that abuts on one side of an analytic curve. The Schwarz reflection principle asserts that under certain conditions, the analytic function extends analytically across the curve, and further, there is a formula for the extension. The formula reflects the function analytically to the mirror image of the domain, and the nub of the problem is to establish analyticity on the axis of reflection.

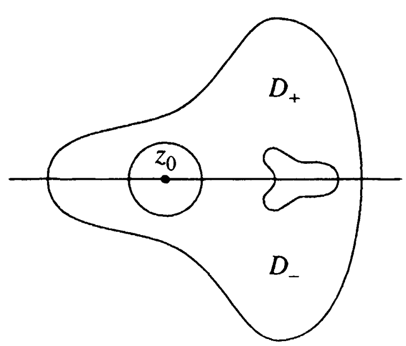
At its core, the Schwarz reflection principle is a theorem about harmonic functions. We begin by reflecting harmonic functions across the real line.

First note that if $u(z)$ is harmonic in a domain $D$, then $u^*(z) = u(\bar{z})$ is harmonic on the reflected domain $D^* = \{\bar{z} : z \in D\}$. Indeed, if $u(x, y)$ has the mean value property on circles, then so does $u(x, -y)$. Another way to see this is to note that the change of variables $(x, y) \mapsto (x, -y)$ does not change the Laplacian operator $\Delta$.

**Theorem.** *Let $D$ be a domain that is symmetric with respect to the real axis, and let $D^+ = D \cap \{\mathrm{Im}\, z > 0\}$ be the part of $D$ in the open upper half-plane. Let $u(z)$ be a real-valued harmonic function on $D^+$ such that $u(z) \to 0$ as $z \in D^+$ tends to any point of $D \cap \mathbb{R}$. Then $u(z)$ extends to be harmonic on $D$, and the extension satisfies*

$$u(\bar{z}) = -u(z), \qquad z \in D. \tag{3.1}$$

For the proof, note that $D$ is the disjoint union of $D^+ = D \cap \{\mathrm{Im}\, z > 0\}$, its reflection $D^- = D \cap \{\mathrm{Im}\, z < 0\}$, and $D \cap \mathbb{R}$. We extend $u(z)$ to $D$ by defining $u(z) = -u(\bar{z})$ for $z \in D^-$, and setting $u(z) = 0$ for $z \in D \cap \mathbb{R}$. Then $u(z)$ is continuous on $D$, $u(z)$ is harmonic on $D^+$ and on $D^-$, and (3.1) holds. We claim that the extended function has the mean value property. Fix $z_0 \in D$. If $z_0$ is in $D^+$ or in $D^-$, then $u(z)$ has the mean value property for small disks centered at $z_0$, since $u(z)$ is harmonic near $z_0$. If $z_0 \in D \cap \mathbb{R}$, then by (3.1) the average value of $u(z)$ over the top half of a circle centered at $z_0$ cancels the average over the bottom half of the circle, and the average of $u(z)$ over the circle is 0, which coincides with $u(z_0)$. Thus $u(z)$ has the mean value property on $D$, and $u(z)$ is harmonic on $D$.




There is a corresponding result for analytic functions. First note that if $f(z)$ is analytic on a domain in the upper half-plane, then $g(z) = \overline{f(\bar{z})}$ is analytic on the reflected domain in the lower half-plane. This can be checked using the Cauchy-Riemann equations or a power series expansion. We can think of $g(z)$ as the composition of an anticonformal map $z \mapsto \bar{z}$, a conformal map $f(z)$ (except where $f'(z) = 0$), and an anticonformal map $w \mapsto \bar{w}$; hence it is conformal.

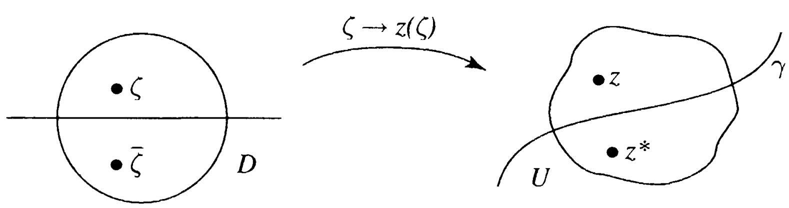
**Theorem.** *Let $D$ be a domain that is symmetric with respect to the real axis, and let $D^+ = D \cap \{\operatorname{Im} z > 0\}$. Let $f(z)$ be an analytic function on $D^+$ such that* $\operatorname{Im} f(z) \to 0$ *as $z \in D^+$ tends to $D \cap \mathbb{R}$. Then $f(z)$ extends to be analytic on $D$, and the extension satisfies*

$$f(\bar{z}) = \overline{f(z)}, \qquad z \in D. \tag{3.2}$$

For the proof, write $f(z) = u(z)+iv(z)$ in $D^+$. By the preceding theorem, $v(z)$ extends to be harmonic on $D$ and satisfies $v(\bar{z}) = -v(z)$, $z \in D$. Fix a point $x_0 \in D \cap \mathbb{R}$, and let $D_0$ be a disk in $D$ centered at $x_0$. Since $v(z)$ is harmonic on $D_0$, it has a harmonic conjugate on $D_0$, and consequently $f(z)$ extends to be analytic on $D_0$. The function $\overline{f(\bar{z})}$ is also analytic on $D_0$, and it coincides with $f(z)$ on the real axis, so by the uniqueness principle it coincides with $f(z)$ in $D_0$. Hence (3.2) holds for $z \in D_0$. If we now extend $f(z)$ to $D^- = D \cap \{\operatorname{Im} z < 0\}$ by setting $f(z) = \overline{f(\bar{z})}$, $z \in D^-$, the extended $f(z)$ is analytic on $D^-$ and coincides with the analytic continuation of $f(z)$ across $D \cap \mathbb{R}$ from $D^+$, and so is analytic on all of $D$. Further, it satisfies (3.2) for all $z \in D$.

The above theorem is somewhat easier to prove if it is assumed that $f(z)$ extends continuously from $D^+$ to $D \cap \mathbb{R}$ and is real-valued there. In this case, formula (3.2) defines a continuous extension of $f(z)$ to all of $D$, which is analytic on $D^-$, and we can use Morera's theorem to show that it is also analytic across $D \cap \mathbb{R}$.

We define a curve $\gamma$ to be an **analytic curve** if every point of $\gamma$ has an open neighborhood $U$ for which there is a conformal map $\zeta \mapsto z(\zeta)$ of a disk $D$ centered on the real line $\mathbb{R}$ onto $U$, such that the image of $D \cap \mathbb{R}$ coincides with $U \cap \gamma$. We also refer to such a $\gamma$ as an **analytic arc**. The conjugation $\zeta \mapsto \bar{\zeta}$ induces a map $z \mapsto z^*$ of $U$ onto itself, by $z(\zeta)^* = z(\bar{\zeta})$. The map is an "involution" in the sense that the map followed by itself is the identity, $z^{**} = z$. Further, $z^* = z$ if and only if $z \in \gamma$. The map $\zeta \mapsto \bar{\zeta}$ interchanges the top half and the bottom half of $D \backslash \mathbb{R}$, which are the two components of $D \backslash \mathbb{R}$, so the map $z \mapsto z^*$ interchanges the two components of $U \backslash \gamma$. We refer to these two components as the **neighborhoods of the sides of** $\gamma$, and we refer to the map $z \mapsto z^*$ as the **reflection across** $\gamma$. Since the map $\zeta \mapsto \bar{\zeta}$ is anticonformal, and the composition of an anticonformal map and conformal maps is anticonformal, the reflection $z \mapsto z^*$ is anticonformal.



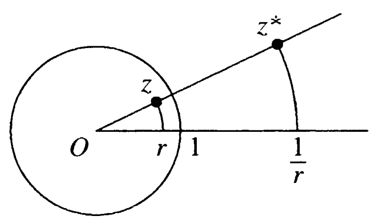

The reflection $z \mapsto z^*$ is unique, in the following sense. If there is an open neighborhood of a point $z_0 \in \gamma \cap U$ and an anticonformal map $z \mapsto z^+$

that fixes each point of $\gamma$ in the neighborhood, then $z^+ = z^*$. Indeed, the map $z \mapsto (z^+)^*$ is the composition of two anticonformal maps, so it is conformal, hence analytic (Section V.8). Since it fixes points of $\gamma$, the uniqueness principle shows that it is the identity map. Thus $(z^+)^* = z$, and $z^+ = (z^+)^{**} = (z^{+*})^* = z^*$.

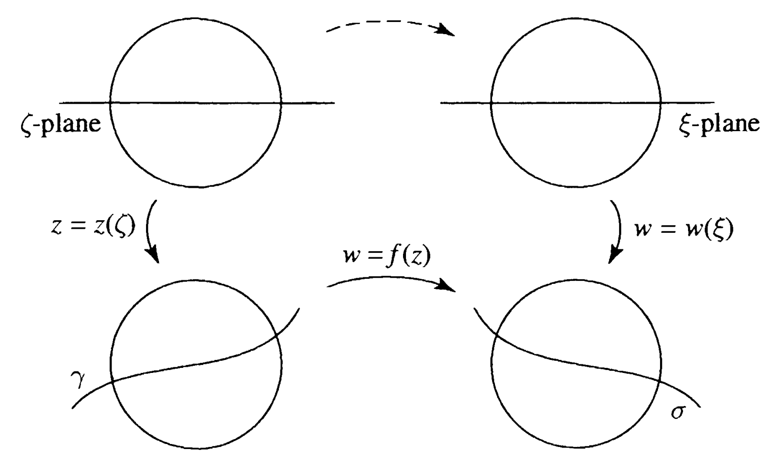
We can extend the involution $z \mapsto z^*$ to a neighborhood of the entire analytic arc $\gamma$, by covering $\gamma$ with small coordinate sets $U$ and using the uniqueness principle to see that the involutions agree on overlapping $U$'s.

**Example.** The map $z(\zeta) = (\zeta - i)/(\zeta + i)$ sends the extended real line onto the unit circle. It induces a reflection across the unit circle,

$$z^* \;=\; (\bar{\zeta} - i)/(\bar{\zeta} + i) \;=\; \overline{(\zeta + i)/(\zeta - i)} \;=\; 1/\bar{z} \;=\; z/|z|^2.$$




From this formula we see that the reflected point $z^*$ lies on the same ray through the origin as $z$. The reflection $z \mapsto z^*$ maps the circle centered at 0 of radius $r$ to that of radius $1/r$, sending $re^{i\theta}$ to $e^{i\theta}/r$.

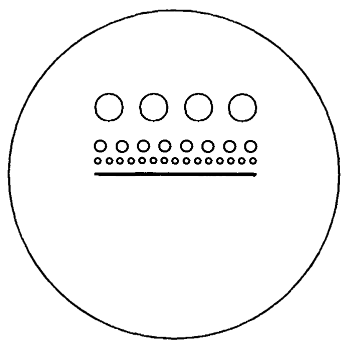



A more general version of the Schwarz reflection principle permits us to reflect harmonic and analytic functions across analytic curves. The idea is that if $w = f(z)$ is defined and analytic on one side of an analytic arc $\gamma$, and if the values $w$ tend to another analytic arc $\sigma$ as $z$ tends to $\gamma$, then $f(z)$ extends analytically across $\gamma$, and the extension satisfies $f(z^*) = f(z)^*$ in a neighborhood of $\gamma$. (Here we use the same notation for both reflections.) It is a straightforward matter to justify the general version by coordinatizing both analytic arcs and reducing to the case of reflection across the real line.

Rather than state a precise theorem, we derive a version of the principle that will be used later.

Let $D$ be a domain. An analytic arc $\gamma \subset \partial D$ is a **free analytic boundary arc** of $D$ if every point of $\gamma$ is contained in a disk $U$ such that $U \backslash \gamma$ has two components, one contained in $D$ and the other disjoint from $D$.

**Example.** Suppose $D$ is obtained from the open disk $\{|z| < 2\}$ by excising a sequence of disjoint closed disks from the top half accumulating on the interval $[-1, 1]$, as in the figure. Then the boundary circle of each excised disk is a free analytic boundary arc of $D$, but the analytic arc $(-1, 1)$ is not a free analytic boundary arc.



**Theorem.** *Let $D$ be a domain, and let $\gamma$ be a free analytic boundary arc of $D$. Let $f(z)$ be analytic on $D$. If $|f(z)| \to 1$ as $z \in D$ tends to $\gamma$, then $f(z)$ extends to be analytic in a neighborhood of $\gamma$, and the extension satisfies*

$$f(z^*) = 1/\overline{f(z)} \tag{3.3}$$

*in a neighborhood of $\gamma$, where $z \mapsto z^*$ is the reflection across $\gamma$.*

For the proof, fix a point $z_0 \in \gamma$, and consider a conformal map $\zeta \mapsto z(\zeta)$ of a disk $D_0$ centered on $\mathbb{R}$ onto an open set containing $z_0$ so that $D_0 \cap \mathbb{R}$ corresponds to $U \cap \gamma$. Then $g(\zeta) = \log f(z(\zeta))$ is defined and analytic on one side of $D_0 \cap \mathbb{R}$. Further, $\operatorname{Re} g(\zeta) = \log |f(z(\zeta))| \to 0$ as $\zeta \to D_0 \cap \mathbb{R}$. By the Schwarz reflection principle, $g(\zeta)$ extends to be analytic on $D_0$, as does $f(z(\zeta)) = e^{g(\zeta)}$. Consequently, $f(z)$ extends analytically to $U$, and the extension is unique. To see that the extension satisfies (3.3), we argue as follows. The function $h(z) = 1/\overline{f(z^*)}$ is the composition of $f(z)$ and the two anticonformal maps $z \mapsto z^*$ and $w \mapsto 1/\bar{w}$. Thus it is conformal wherever $f'(z) \neq 0$. By the equivalence of analyticity and conformality (Section IV.8), $h(z)$ is analytic wherever $f'(z) \neq 0$, and it is continuous, hence analytic also at the isolated points where $f'(z) = 0$. Further, if $z \in \gamma$, then $z^* = z$, and $|f(z)| = 1$, so $h(z) = f(z)$ on $\gamma$. By the uniqueness principle, $h(z) = f(z)$ everywhere, and $f(z)$ satisfies (3.3).

## Exercises for X.3

1. Show that the reflection in the circle $\{|z - z_0| = R\}$ is given by $z^* = z_0 + R^2(z - z_0)/|z - z_0|^2$.

2. Show that a reflection in a circle maps circles in the plane to circles.

3. What happens to angles between curves when they are reflected in an analytic arc?

4. Suppose the curve $\gamma$ passing through 0 is the graph of a function $y = h(x)$ that can be expressed as a convergent power series $h(x) = \sum_{k=1}^{\infty} a_k x^k$, $-r < x < r$, where the $a_k$'s are real. (a) Show that $z = \zeta + ih(\zeta)$ can be solved for $\zeta = \zeta(z)$ as an analytic function of $z$ for $|z| < \varepsilon$. (b) Show that $\gamma$ is an analytic arc. (c) Show that the reflection through $\gamma$ is given by $z^* = 2\overline{\zeta(z)} - \bar{z}$.

5. Determine the reflection $z^*$ of $z$ across the parabola $y = x^2$. Expand $z^*$ in a power series, in powers of $\bar{z}$. Determine the radius of convergence of the series. Try to explain graphically why the radius of convergence is finite.

6. Let $f(z)$ be an entire function whose modulus is constant on some circle. Show that $f(z) = c(z - z_0)^n$ for some $n \geq 0$ and some constant $c$, where $z_0$ is the center of the circle.

7. Show that if $f(z)$ is meromorphic for $|z| < 1$, and $|f(z)| \to 1$ as $|z| \to 1$, then $f(z)$ is a rational function. Show further that $f(z)$ is the quotient of two finite Blaschke products. (For the definition of finite Blaschke product, see Exercise IX.2.3.)

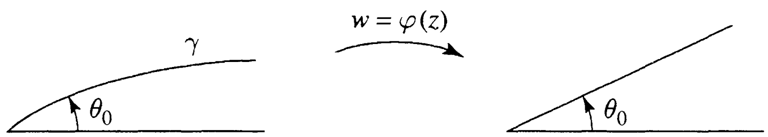
8. The **modulus of an annulus** $\{a < |z - z_0| < b\}$ is defined to be $(1/2\pi)\log(b/a)$. (a) Show that any conformal map from one annulus centered on the origin to another such annulus extends to a conformal self-map of the punctured plane. (b) Show that there is a conformal map of one annulus onto another if and only if the annuli have the same moduli. (c) Show that any conformal self-map of the annulus $\{a < |z| < b\}$ is either a rotation $z \mapsto e^{i\varphi}z$ or a rotation followed by the inversion $z \mapsto ab/z$.

9. Let $\gamma$ be an analytic curve passing through the origin and tangent to the imaginary axis at 0. Suppose that $\gamma(t) = it + b_2 t^2 + b_3 t^3 + \cdots$, $-\delta < t < \delta$, where the $b_n$'s are real. Show that there is a conformal map $\varphi(z)$ defined in a neighborhood of 0 such that $\varphi(0) = 0$, $\varphi'(0) = 1$, and $\varphi(z)$ maps the angle between the positive real axis and the segment of $\gamma$ in the upper half-plane to the angle between the positive real and imaginary axes if and only if $\gamma$ coincides with its reflection $\bar{\gamma}$ in the real axis. Show that this occurs if and only if $b_n = 0$ for $n$ odd.

10. Let $\gamma$ be an analytic curve passing through the origin and making an angle $\theta_0$, $0 < \theta_0 < \pi/2$, with the positive real axis. Suppose $\gamma$ is the image of an interval $(-\delta, \delta)$ under $h(z) = z + i(b_1 z + b_2 z^2 + \cdots)$, where the $b_j$'s are real. Define

$$g(z) = \overline{h(\,\overline{h^{-1}(z)}\,)}, \qquad \lambda = e^{-2i\theta_0}.$$

(a) Show that $g(z)$ is analytic at $z = 0$, $g'(0) = \lambda$, and $g(z) = \bar{z}$ for $z \in \gamma$.




(b) Show that if $\varphi(z)$ is analytic at $z = 0$, $\varphi(0) = 0$, $\varphi'(0) = 1$, and $\varphi(z)$ maps the angle between the positive real axis and the segment of $\gamma$ in the first quadrant to the angle between the positive real axis and the straight line segment at angle $\theta_0$, then $\varphi(g(z)) = \lambda\varphi(z)$. *Remark.* This equation for $\varphi(z)$ is **Schröder's equation**.

(c) Show by plugging in power series that any (normalized) solution $\varphi(z)$ of Schröder's equation is unique, if it exists, and is given by $\varphi(z) = z + c_2 z^2 + c_3 z^3 + \cdots$, where

$$c_n = \frac{1}{\lambda^n - \lambda} A_n(\lambda, a_2, \ldots, a_n, c_2, \ldots, c_{n-1}), \qquad n \geq 2,$$

$A_n$ is a polynomial, and the $a_n$'s are the power series coefficients of $g(z)$. *Remark.* The problem of estimating the $c_n$'s to determine whether the series converges is called a "small denominator problem ." As the powers $\lambda^{n-1}$ return sporadically near to 1, the denominators become sporadically small.

(d) Establish the converse of (b) by showing that any solution $\varphi(z)$ of Schröder's equation (analytic at 0 and satisfying $\varphi(0) = 0$, $\varphi'(0) = 1$) maps the angle between the positive real axis and $\gamma$ to the angle between the straight line segments at angles 0 and $\theta_0$.

# XI

# Conformal Mapping

In this chapter we will be concerned with conformal maps from domains onto the open unit disk. One of our goals is the celebrated Riemann mapping theorem: Any simply connected domain in the complex plane, except the entire complex plane itself, can be mapped conformally onto the open unit disk. We begin in Section 1 by reviewing and enlarging our repertoire of conformal maps onto the open unit disk, or equivalently, onto the upper half-plane. In Section 2 we state and discuss the Riemann mapping theorem. Before embarking on the proof, we give some applications to the conformal mapping of polygons in Section 3 and to fluid dynamics in Section 4. In Section 5 we develop some prerequisite material concerning compactness of families of analytic functions, which is at a deeper level than the analysis used up to this point. The proof of the Riemann mapping theorem follows in Section 6.

## 1. Mappings to the Unit Disk and Upper Half-Plane

Recall that a **conformal map** of a domain $D$ onto a domain $V$ is an analytic function $\varphi(z)$ from $D$ to $V$ that is one-to-one and onto. The composition of two conformal maps is a conformal map, and the inverse of a conformal map is a conformal map. In this section we seek to find conformal maps from various domains $D$ onto the open unit disk $\mathbb{D}$. Such a map $\varphi(z)$ is never unique, as we can compose it with any conformal self-map of $\mathbb{D}$. However, if $\psi(z)$ is any other conformal map of $D$ onto $\mathbb{D}$, then $g = \psi \circ \varphi^{-1}$ is a conformal self-map of $\mathbb{D}$ for which $g \circ \varphi = \psi$. Thus once one conformal map $\varphi$ of $D$ onto $\mathbb{D}$ is known, the others are precisely the maps of the form $\psi = g \circ \varphi$, where $g$ is a conformal self-map of $\mathbb{D}$.

We have seen in Section IX.2 that the conformal self-maps of the open unit disk have the form

$$g(z) = \lambda \frac{z-a}{1-\bar{a}z}, \qquad z \in \mathbb{D},$$

where $|a| < 1$ and $|\lambda| = 1$. The conformal self-maps of $\mathbb{D}$ are determined by three real parameters, namely, the real and imaginary parts of $a$, and the argument of the unimodular constant $\lambda$. Consequently, a conformal map of $D$ onto $\mathbb{D}$ is uniquely determined by specifying three real parameters. Specifying the image of a fixed point of $D$ counts as two real parameters, corresponding to real and imaginary parts. Specifying the image of a boundary point of $D$ counts as one real parameter.

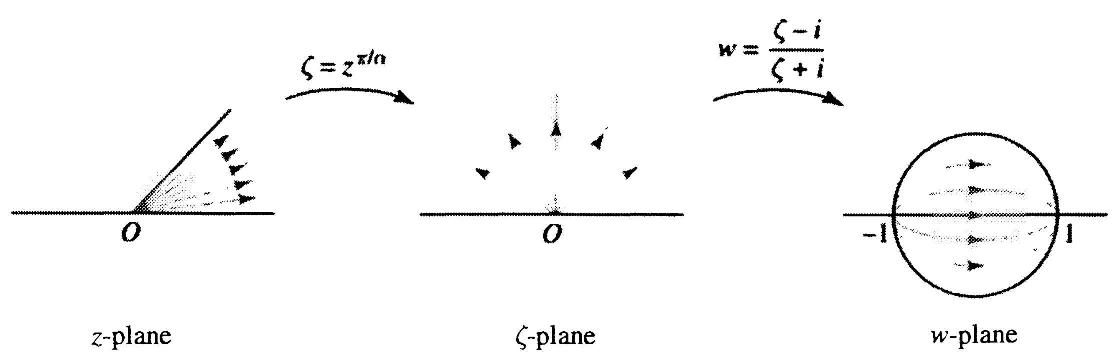
The fractional linear transformation $w = (z - i)/(z + i)$ maps the open upper half-plane $\mathbb{H}$ onto the open unit disk $\mathbb{D}$. The problem of finding a conformal map of a given domain onto $\mathbb{D}$ is equivalent to finding one onto $\mathbb{H}$. We can go back and forth between $\mathbb{D}$ and $\mathbb{H}$ by composing with the above map and its inverse $z = i(1 + w)/(1 - w)$. The complex analyst does not distinguish between the half-plane and the unit disk, and works in whichever space the calculations are simpler.

We turn to three specific classes of domains: sectors, strips, and lunes.

**Sectors.** A sector can be mapped onto a half-plane with the aid of the power function $z^\beta$, for an appropriate choice of $\beta$. Any sector with vertex at 0 can be rotated by the map $z \mapsto \lambda z$, $|\lambda| = 1$, to a sector of the form $D = \{0 < \arg z < \alpha\}$, where $\alpha \le 2\pi$. Since $z^\beta$ multiplies angles by $\beta$, we set $\beta = \pi/\alpha$, to obtain a map $\zeta = z^{\pi/\alpha}$ of the sector $D$ onto the upper half-plane. If we follow this by the map $w = (\zeta - i)/(\zeta + i)$ of $\mathbb{H}$ onto $\mathbb{D}$, we obtain a conformal map

$$w = \varphi(z) = \frac{z^{\pi/\alpha} - i}{z^{\pi/\alpha} + i}, \qquad z \in D,$$

mapping the sector $D$ onto the open unit disk. Under this map, the vertex of the sector at $z = 0$ corresponds to $w = -1$, and the other "vertex" at $z = \infty$ corresponds to $w = +1$. Rays emanating from the origin are mapped to arcs of circles from $-1$ to $+1$ in $\mathbb{D}$, and the circular arcs $\{|z| = \text{ constant}\}$ are mapped to their orthogonal trajectories, which are arcs of circles from the bottom half of the unit circle to the top half.




**Exercise.** Find a conformal map of the sector $\{-\pi < \arg z < \pi/2\}$ onto the open unit disk.

**Solution.** The rotation $\zeta = e^{\pi i} z = -z$ maps the given sector to the sector $\{0 < \arg\zeta < 3\pi/2\}$ in the $\zeta$-plane. The power function $\xi = \zeta^{2/3}$ maps this to the upper half of the $\xi$-plane. We compose this with the map $w = (\xi - i)/(\xi + i)$ onto the open unit disk, and we obtain

$$w = \frac{(-z)^{2/3} - i}{(-z)^{2/3} + i} = \frac{e^{2\pi i/3}z^{2/3} - i}{e^{2\pi i/3}z^{2/3} + i},$$

where we take the branch of $z^{2/3}$ satisfying $-2\pi/3 < \arg\left(z^{2/3}\right) < \pi/3$ on the sector.

**Strips.** An arbitrary strip can be mapped to a horizontal strip by a rotation $z \mapsto \lambda z$. The exponential function $e^z$ maps horizontal strips to sectors. Thus an appropriate power $e^{\alpha z}$ maps a horizontal strip onto a half-plane. A further rotation maps the half-plane onto the upper half-plane, which can be then mapped to the unit disk.

**Exercise.** Find a conformal map of the vertical strip $\{-1 < \operatorname{Re} z < 1\}$ onto the open unit disk $\mathbb{D}$. What are the images of vertical and horizontal lines in the strip under the map?
**Solution.** The preliminary rotation $\zeta = iz$ maps the vertical strip to the horizontal strip $\{-1 < \operatorname{Im} z < 1\}$. The exponential map $\xi = e^{\alpha\zeta}$ maps the horizontal strip onto the sector $\{-\alpha < \arg\xi < \alpha\}$. Thus if we take $\alpha = \pi/2$, we map onto the right half of the $\xi$-plane. Multiplication by $i$ maps this onto the upper half-plane, and we obtain finally

$$w = \frac{i\xi - i}{i\xi + i} = \frac{\xi - 1}{\xi + 1} = \frac{e^{\pi\zeta/2} - 1}{e^{\pi\zeta/2} + 1} = \frac{e^{\pi iz/2} - 1}{e^{\pi iz/2} + 1}.$$

Under this map, $z = -i\infty$ is mapped to $w = +1$ and $z = +i\infty$ is mapped to $w = -1$. Vertical lines in the strip in the $z$-plane are rotated to horizontal lines in the $\zeta$-plane, which are carried by the exponential map to rays in the $\xi$-plane. These are carried to circles in the unit disk in the $w$-plane passing through $\pm 1$. Horizontal lines in the strip are mapped to the orthogonal trajectories of the circles passing through $\pm 1$. These orthogonal trajectories are also arcs of circles, passing from the bottom half of the unit circle to the top half.

Note that the solution map for this problem is not unique, and different maps will give different images of horizontal and vertical lines. However, once the images of $-i\infty$ and $+i\infty$ are known, then the vertical lines in the strip will be mapped to arcs of circles through these two image points, and horizontal lines in the strip will be mapped to the arcs of circles that are their orthogonal trajectories.

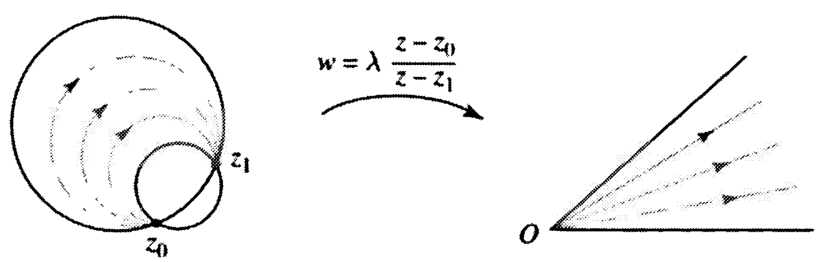
**Exercise.** Find a conformal map of the vertical strip $\{-1 < \operatorname{Re} z < 1\}$ onto the open unit disk $\mathbb{D}$ that maps $-i\infty$ to $-1$ and $+i\infty$ to $i$.
**Solution.** The desired map is obtained by composing the map just constructed with a conformal self-map $g(w) = \lambda(w - a)/(1 - \bar{a}w)$ of the unit

disk. Since $z = -i\infty$ goes to $w = +1$, we require that $g(+1) = -1$, and since $z = i\infty$ goes to $w = -1$, we require that $g(-1) = i$. Thus

$$\lambda \frac{1-a}{1-\bar{a}} = -1, \qquad \lambda \frac{-1-a}{1+\bar{a}} = i.$$

If we eliminate the unimodular constant $\lambda$ and simplify, we obtain $|a|^2 + (\bar{a} - a) = 1$, which is the equation of a circle centered at $+1$ of radius $\sqrt{2}$. Thus the solution is not unique, as we can choose $a$ to be any point on this circle satisfying $|a| < 1$. For instance, we could take $a = 1 - \sqrt{2}$, which yields $\lambda = -1$. Choosing the point $a$ on the circle corresponds to specifying the third parameter. Note that there is no solution for which additionally the point $z = 0$ is mapped to 0. We would ordinarily not expect there to be such a map, since this would correspond to specifying a total of four parameters, and we have only three at our disposal.

**Lunar domains.** Suppose the domain $D$ has boundary consisting of two curves, each of which is an arc of a circle or a straight line segment. Let $z_0$ and $z_1$ be the endpoints of the curves. We assume $z_0 \neq z_1$. Then there is a fractional linear transformation $\zeta = g(z)$ mapping $z_0$ to 0 and $z_1$ to $\infty$. Since fractional linear transformations map circles to circles, the images of the two arcs lie on circles passing through 0 and $\infty$, and so are rays from 0 to $\infty$. Consequently, $\zeta = g(z)$ maps $D$ onto a sector in the $\zeta$-plane. This reduces the problem to mapping the sector onto $\mathbb{H}$ or $\mathbb{D}$.

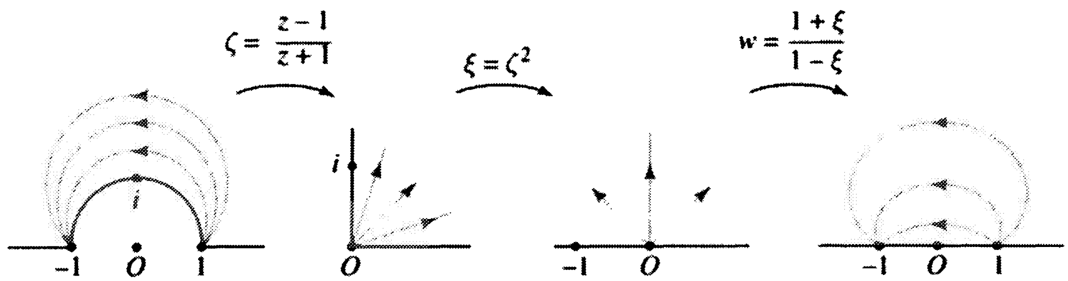



**Exercise.** Find a conformal map of the part of the upper half-plane outside the unit circle onto the entire upper half-plane mapping $-1$ to $-1$, $i$ to 0, and $+1$ to $+1$.

**Solution.** We think of $D$ as a lunar domain in the extended complex plane. One of the bounding arcs is the straight line segment from $+1$ through $\infty$ to $-1$, and the other bounding arc is the top half of the unit circle. The map $\zeta(z) = (z-1)/(z+1)$ sends the two vertices $-1$ and $+1$ of the lune to $\infty$ and 0, respectively, and it maps $D$ to the first quadrant. Under this map, $i$ goes to $i$. The first quadrant is mapped to the upper half-plane by $\xi = \zeta^2$. Thus the composition $\xi = (z-1)^2/(z+1)^2$ maps $D$ onto the upper half-plane, and it sends $-1$ to $\infty$, $i$ to $-1$, and $+1$ to 0. Finally, we compose with the fractional linear transformation that maps the upper half-plane to itself and sends $\infty$ to $-1$, $-1$ to 0, and 0 to 1. This map is found to be $w = (1+\xi)/(1-\xi)$. The final solution is then

$$(1.1)\quad w = \frac{1+(z-1)^2/(z+1)^2}{1-(z-1)^2/(z+1)^2} = \frac{(z+1)^2+(z-1)^2}{(z+1)^2-(z-1)^2} = \frac{1}{2}\left(z+\frac{1}{z}\right),$$

which has already appeared in Exercise II.6.6.




## Exercises for XI.1

1. Find a conformal map of the sector $\{|\arg z| < \pi/3\}$ onto the open unit disk mapping 0 to $-1$ and $\infty$ to $+1$. Sketch the images of radial lines and of arcs of circles centered at 0. Is the map unique?

2. Find a conformal map of the slit plane $\mathbb{C}\backslash(-\infty, 0]$ onto the open unit disk satisfying $w(0) = i$, $w(-1+0i) = +1$, $w(-1-0i) = -1$. What are the images of circles centered at 0 under the map? Sketch them.

3. For fixed $A > 0$, find the conformal map $w(z)$ of the open unit disk $\{|z| < 1\}$ onto the vertical strip $\{-A < \operatorname{Re} w < A\}$ that satisfies $w(0) = 0$ and $w'(0) > 0$. Sketch the curves in the disk that correspond to vertical and horizontal lines in the strip.

4. Find a conformal map $w(z)$ of the strip $\{\operatorname{Im} z < \operatorname{Re} z < \operatorname{Im} z + 2\}$ onto the upper half-plane such that $w(0) = 0$, $w(z) \to +1$ as $\operatorname{Re} z \to -\infty$, and $w(z) \to -1$ as $\operatorname{Re} z \to +\infty$. Sketch the images of the straight lines $\{\operatorname{Re} z = \operatorname{Im} z + c\}$ in the strip. What is the image of the median line $\{\operatorname{Re} z = \operatorname{Im} z + 1\}$ of the strip?

5. Find a conformal map $w(z)$ of the right half-disk $\{\operatorname{Re} z > 0, |z| < 1\}$ onto the upper half-plane that maps $-i$ to 0, $+i$ to $\infty$, and 0 to $-1$. What is $w(1)$?

6. Let $w = g(z)$ be the conformal map of the right half-disk $\{\operatorname{Re} z > 0, |z| < 1\}$ onto the entire unit disk that fixes the points $\pm i$ and $+1$. (a) Without computing $g(z)$ explicitly, show that $g(\bar{z}) = \overline{g(z)}$. *Hint.* Argue that $h(z) = \overline{g(\bar{z})}$ is another conformal map satisfying the same conditions, and appeal to uniqueness. (b) Use symmetry to show that $g(0) = -1$. (In other words, use part (a).) (c) Find

$g(z)$ as a composition of explicit conformal maps, and use this to check that $g(0) = -1$.

7. Find the conformal map of the pie-slice domain $\{|\arg z| < \alpha, |z| < 1\}$ onto the open unit disk such that $w(0) = -1$, $w(+1) = +1$, and $w(e^{i\alpha}) = i$. It is enough to express $w(z)$ as a composition of specific conformal maps.

8. For fixed $b$ in the interval $(-1, 1)$, find all conformal maps of the unit disk slit along the interval $[-1, b]$ onto the entire unit disk that map $b$ to $-1$ and leave $+1$ fixed. It is enough to express them as a composition of specific conformal maps.

9. For fixed $a > 0$, let $D$ be the domain obtained by slitting the upper half-plane along the vertical interval from $z = 0$ to $z = ia$.
   (a) Find a conformal map $w(z)$ of $D$ onto the entire upper half-plane such that $w(z) \sim z$ as $|z| \to \infty$. *Hint.* Consider the preliminary map $\zeta = z^2$.
   (b) Describe how the map can be used to model the flow of water over a vertical metal sheet lying in a flat river bed, perpendicular to the flow of the water. Give a rough sketch of the streamlines of the flow.

10. Find the conformal map $w = f(z)$ of the exterior domain $\{|z| > 1\} \cup \{\infty\}$ onto $\mathbb{C}^* \backslash [-1, +1]$ such that $f(\infty) = \infty$ and the argument of $f(z)/z$ tends to $\alpha$ as $z \to \infty$, where $\alpha$ is a fixed real number. Sketch the images of circles $\{|z| = r\}$, for $r > 1$, and the images of the intervals $(-\infty, -1]$ and $[1, \infty)$. What is the inverse map? (Specify the branch.) *Hint.* For $\alpha = 0$, use the map $(z + 1/z)/2$ treated in (1.1) above. For the general case, do a preliminary rotation. See also the exercises in Section II.6.

11. Show that the half-strip $\{-\pi/2 < \operatorname{Re} z < \pi/2, \operatorname{Im} z > 0\}$ is mapped conformally by $w = \sin z$ onto the upper half-plane. Sketch the images of horizontal and vertical lines.

12. Find a conformal map of the half-strip in Exercise 11 onto the open unit disk that maps $-\pi/2$ to $-i$, $\pi/2$ to $i$, and $0$ to $+1$. Where does $\infty$ go under this map?

## 2. The Riemann Mapping Theorem

This section is devoted to a preliminary discussion of the Riemann mapping theorem. The proof will be postponed to the end of the chapter. The Riemann mapping theorem was first established in the following generality by W.F. Osgood in 1900.

**Theorem (Riemann Mapping Theorem).** *If $D$ is a simply connected domain in the complex plane, and $D$ is not the entire complex plane, then there is a conformal map of $D$ onto the open unit disk $\mathbb{D}$.*

Concerning the hypothesis, recall from Section VIII.8 that a domain is simply connected if every closed curve in the domain can be deformed to a point in the domain. Several characterizations of simply connected domains were given in Section VIII.8. Roughly speaking, a domain is simply connected if the complement has no "holes." Disks are simply connected, and annuli are not.

We say that two domains are **conformally equivalent** if there is a conformal map of one onto the other. Thus the Riemann mapping theorem asserts that any simply connected domain in the complex plane $\mathbb{C}$ either coincides with $\mathbb{C}$ or is conformally equivalent to $\mathbb{D}$.

We refer to a conformal map $w = \varphi(z)$ of $D$ onto $\mathbb{D}$ as the **Riemann map** of $D$ onto $\mathbb{D}$. It is unique, up to postcomposing with a conformal self-map of $\mathbb{D}$. To specify the Riemann map uniquely we must specify three real parameters. One way to do this is to specify $\varphi(z_0) = 0$ and $\varphi'(z_0) > 0$ for some $z_0 \in D$.

Each of the hypotheses in the Riemann mapping theorem is necessary. The complex plane $\mathbb{C}$ cannot be mapped conformally onto any bounded domain, because according to Liouville's theorem the only bounded analytic functions on $\mathbb{C}$ are the constants. Since any closed curve in $\mathbb{D}$ can be deformed to a point, any closed curve in any domain conformally equivalent to $\mathbb{D}$ can also be deformed to a point, by composing a deformation in $\mathbb{D}$ with the inverse of the Riemann map. Thus any domain conformally equivalent to $\mathbb{D}$ is simply connected.

For a simply connected domain $D$ in the Riemann sphere (the extended complex plane), there are three possibilities. Indeed, if the domain is not the entire Riemann sphere, we can move a point in the complement of the domain to $\infty$ by a fractional linear transformation and reduce to the case of a simply connected domain in the plane, where there are two possibilities. The Riemann mapping theorem then yields the following.

**Corollary.** *A simply connected domain in the Riemann sphere is either the entire Riemann sphere, or it is conformally equivalent to the complex plane, or it is conformally equivalent to the open unit disk.*

Suppose now that $D$ is a simply connected domain with Riemann map $\varphi(z)$ mapping $D$ onto $\mathbb{D}$. We consider the behavior of $\varphi(z)$ as $z$ approaches the boundary of $D$. For each fixed $\varepsilon > 0$, the set $\{|\varphi(z)| \leq 1 - \varepsilon\}$ is a compact subset of $D$, which is at a positive distance from $\partial D$. It follows that $|\varphi(z)| \to 1$ as $z \to \partial D$. Hence the harmonic function $\log|\varphi(z)|$ tends to 0 as $z$ tends to the boundary of $D$. The Schwarz reflection principle for harmonic functions (Section X.3) then implies that $\log|\varphi(z)|$ extends harmonically across any free analytic boundary arc of $D$. Consequently, $\varphi(z)$ extends analytically across any free analytic boundary arc of $D$, and in fact we have the following.

**Theorem.** *Let $D$ be a simply connected domain in $\mathbb{C}$, $D \neq \mathbb{C}$. Then the Riemann map $\varphi(z)$ of $D$ onto $\mathbb{D}$ extends analytically across any free analytic boundary arc $\gamma$ of $D$, and $\varphi(z)$ maps $\gamma$ one-to-one onto an arc of $\partial\mathbb{D}$. The extended function satisfies $\varphi'(z) \neq 0$ for $z \in \gamma$, and $\varphi(z^*) = 1/\overline{\varphi(z)}$ for $z$ in a neighborhood of $\gamma$, where $z \mapsto z^*$ is the reflection across $\gamma$. Disjoint free analytic boundary arcs of $D$ are mapped by $\varphi(z)$ to disjoint arcs of $\partial\mathbb{D}$.*

For the formula for the analytic extension, see Section X.3. From the formula and the fact that $|\varphi(z)| < 1$ on the side of $\gamma$ in $D$, we see that $|\varphi(z)| > 1$ on other side of $\gamma$. Thus near any point of $\gamma$ the locus where $|\varphi(z)| = 1$ consists only of points of the one analytic curve $\gamma$, so no point of $\gamma$ can be a critical point of $\varphi(z)$. Let $z_0 \in \gamma$, and let $\zeta_0 = \varphi(z_0)$. If $U_\varepsilon$ is a small disk centered at $z_0$, then $\varphi(U_\varepsilon)$ includes all points of $\mathbb{D}$ in a sufficiently small disk centered at $\zeta_0$. Since $\varphi(z)$ is one-to-one on $D$, no other point of $D$ is mapped by $\varphi(z)$ to this disk centered at $\zeta_0$. Thus no other point of $\gamma$, and no point lying on any other free analytic boundary arc of $D$, is mapped to $\zeta_0$. It follows that $\varphi(z)$ is one-to-one on $\gamma$, and further, the image of $\gamma$ is disjoint from the image of any other free analytic boundary arc of $D$.

### Exercises for XI.2

1. Show that no two of the domains $\mathbb{C}^*$, $\mathbb{C}$, and $\mathbb{D}$ are conformally equivalent.

2. Let $\varphi(z)$ be a conformal map from a domain $D$ onto the open unit disk $\mathbb{D}$. For $0 < r < 1$, let $D_r$ be the set of $z \in D$ such that $|\varphi(z)| < r$. Find a conformal map of $D_r$ onto $\mathbb{D}$.

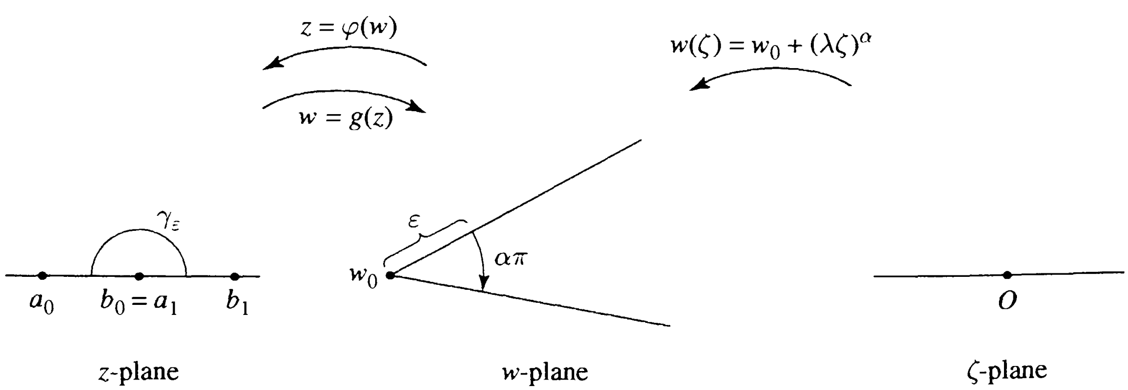
3. Let $D$ be a domain in the complex plane whose complement $\mathbb{C}^*\backslash D$ in the extended complex plane consists of a finite number of disjoint closed connected sets, not all of which are points. Show that $D$ can be mapped conformally onto a bounded domain whose boundary consists of a finite number of points and simple closed analytic curves.

## 3. The Schwarz-Christoffel Formula

Suppose that $D$ is a polygonal domain, bounded by a finite number of straight line segments, and $w = g(z)$ is a conformal map of the upper half-plane onto $D$. The Schwarz-Christoffel formula is a differential equation that $g(z)$ must satisfy. Even though the differential equation cannot be solved in closed form except in very special cases, it provides a basis for computation of conformal maps onto arbitrary domains by approximating them by polygons and solving the associated differential equations numerically.

We begin by studying the behavior of a conformal map at a corner. It will be convenient to work in the upper half-plane instead of the unit disk.

Let $D$ be a simply connected domain that has a corner at $w_0$, so that the part of $D$ near $w_0$ has the form $D_\delta = \{w_0 + re^{i\theta} : 0 < r < \delta,\, \theta_1 < \theta < \theta_0\}$. Thus the part of $\partial D$ near $w_0$ consists of two straight line segments terminating at $w_0$ and subtending an interior angle of $\theta_0 - \theta_1 = \pi\alpha$, where $0 < \alpha < 2$. Let $z = \varphi(w)$ be the Riemann map of $D$ onto the upper half-plane $\mathbb{H}$. It maps the two sides of the corner analytically onto two disjoint intervals on the extended real line. By composing with a conformal self-map of $\mathbb{H}$, we can assume that the side at angle $\theta_0$ is mapped to a finite interval $I = (a_0, b_0)$, and the side at angle $\theta_1$ to a finite interval $J = (a_1, b_1)$, ordered so that $b_0 \le a_1$. The image under $\varphi(w)$ of the circular arc $\{|w - w_0| = \varepsilon\} \cap D$ is a curve $\gamma_\varepsilon$ in $\mathbb{H}$ from $I$ to $J$. Since the circular arcs tend to $w_0 \in \partial D$, the curves $\gamma_\varepsilon$ tend to the boundary of $\mathbb{H}$ as $\varepsilon \to 0$, in such a way that the initial point of $\gamma_\varepsilon$ increases to $b_0$ and the terminal point decreases to $a_1$. It follows that the curves tend to the closed interval $[b_0, a_1]$ as $\varepsilon \to 0$. Thus the inverse function $w = g(z)$ of $\varphi(w)$ tends to the vertex $w_0$ as $z \in \mathbb{H}$ tends to the interval $[b_0, a_1]$. Now, the interval cannot have any interior points, or else $g(z) - w_0$ would tend to zero on the interval, hence reflect analytically across, hence be identically zero, which is absurd. Thus $b_0 = a_1$. We conclude that $\varphi(w)$ extends continuously to the corner $w_0$, and that $\varphi(w)$ maps the two segments in $\partial D$ abutting at a corner $w_0$ to two contiguous intervals on the real line meeting at the image $\varphi(w_0)$ of the corner.




Now we consider more carefully the behavior of the inverse map $w = g(z)$ of the upper half-plane onto the domain $D$, mapping the point $a_0 \in \mathbb{R}$ to the corner $w_0$, and mapping intervals on each side of $a_0$ onto two straight line segments in $\partial D$ terminating at $w_0$. Let $\lambda = e^{i\theta_1}$, a unimodular constant.

The function $w(\zeta) = w_0 + \lambda\zeta^\alpha$ maps a semidisk $\{|\zeta| < \varepsilon, \operatorname{Im}\zeta > 0\}$ onto the part of $D$ at the corner. We may express $\zeta$ as a function of $z$, by $\lambda^{1/\alpha}\zeta(z) = (g(z) - w_0)^{1/\alpha}$. Since $\operatorname{Im}\zeta(z) \to 0$ as $\operatorname{Im} z \to 0$, the Schwarz reflection principle shows that $\zeta(z)$ is analytic at $z = a_0$, and further, $a_0$ cannot be a critical point of $\zeta(z)$. Thus $\lambda^{1/\alpha}\zeta(z) = (z-a_0)f(z)$ where $f(z)$ is analytic at $a_0$ and $f(a_0) \neq 0$. Let $h(z)$ be an analytic branch of $f(z)^\alpha$, so that $\lambda\zeta^\alpha = (z - a_0)^\alpha h(z)$ and $h(a_0) \neq 0$. Then from $g(z) = w_0 + \lambda\zeta^\alpha$ we obtain

$$g(z) = w_0 + (z - a_0)^\alpha h(z),$$

where $h(z)$ is analytic at $a_0$ and $h(a_0) \neq 0$. If we differentiate once, we eliminate $w_0$. If we differentiate again, we obtain

$$\begin{aligned} g'(z) &= (z - a_0)^{\alpha-1}\left(\alpha h(z) + \mathcal{O}(z - a_0)\right), \\ g''(z) &= (z - a_0)^{\alpha-2}\left(\alpha(\alpha - 1)h(z) + \mathcal{O}(z - a_0)\right), \end{aligned}$$

and this allows us to eliminate the branch point by dividing,

$$(3.1) \qquad \frac{g''(z)}{g'(z)} = \frac{\alpha - 1}{z - a_0} + \text{ analytic}, \qquad |z - a_0| < \varepsilon,\ \operatorname{Im} z > 0.$$

Thus while $g(z)$ has a branch point at $a_0$, the combination of derivatives $g''(z)/g'(z)$ extends meromorphically to $a_0$ and has a simple pole there with residue $\alpha - 1$.

Something similar happens if the point $\infty$ is mapped to the corner $w_0$ by $g(z)$. We parametrize the corner by $\zeta$ as before, and apply the Schwarz reflection principle, to see that $\lambda^{1/\alpha}\zeta(z) = (g(z) - w_0)^{1/\alpha}$ depends analytically on $z$ at $z = \infty$. In this case, $\lambda^{1/\alpha}\zeta(z) = f(z)/z$, where $f(z)$ is analytic at $\infty$ and $f(\infty) \neq 0$. Then $g(z) = w_0 + h(z)/z^\alpha$, where $h(z)$ is analytic at $\infty$ and $h(\infty) \neq 0$. A calculation similar to the one above shows that

$$(3.2) \qquad \frac{g''(z)}{g'(z)} = -\frac{2}{z} + \cdots, \qquad |z| > R,\ \operatorname{Im} z > 0,$$

is analytic at $\infty$ and vanishes there. We have proved the following theorem on corners.

**Theorem.** *Let $D$ be a simply connected domain with a corner at $w_0 \in \partial D$ with interior angle $\alpha\pi$, $0 < \alpha < 2$. Suppose $w = g(z)$ is a conformal map of $\mathbb{H}$ onto $D$. Then there are two contiguous intervals of the extended real line that are mapped analytically by $g(z)$ onto the sides of the corner, and $g(z)$ extends continuously to map the common endpoint $a_0$ of the intervals to the vertex $w_0$ of the corner. If $a_0$ is finite, then $g''(z)/g'(z)$ extends to be meromorphic at $a_0$ with a simple pole and with residue $\alpha - 1$. If $a_0 = \infty$, then $g''(z)/g'(z)$ extends to be analytic at $\infty$ and vanishes there.*

The preceding theorem also applies to a virtual corner, that is, to a point lying on straight line segments in $\partial D$. Such a point can be considered a corner with angle $\pi$. If $a_0$ is finite, the residue at $a_0$ is zero, and $g''(z)/g'(z)$ is analytic at $a_0$, as it is also in the case that $a_0 = \infty$.

If $D$ is a polygon, we can carry the analysis further. In this case, the preceding analysis shows that the conformal map $g(z)$ of $\mathbb{H}$ onto $D$ extends continuously to map the extended real line one-to-one onto $\partial D$.

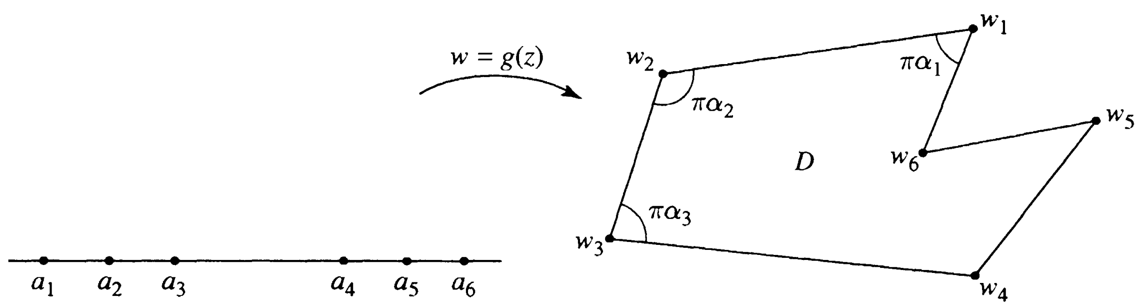
**Theorem.** *Suppose $g(z)$ is a conformal map of the upper half-plane onto a (bounded) polygon $D$. Let $a_1 < \cdots < a_m$ be the points of $\mathbb{R}$ mapped to vertices of $D$. (It may be that $\infty$ is also mapped to a vertex.) Suppose $D$ has interior angle $\alpha_j \pi$ at the vertex $w_j = g(a_j)$, $1 \le j \le m$. Then*

$$\frac{g''(z)}{g'(z)} = \frac{\alpha_1 - 1}{z - a_1} + \cdots + \frac{\alpha_m - 1}{z - a_m}, \tag{3.3}$$

*and there are constants $A$ and $B$ such that*

$$g'(z) = A(z-a_1)^{\alpha_1 - 1} \cdots (z-a_m)^{\alpha_m - 1}, \tag{3.4}$$

$$g(z) = A \int_{z_0}^{z} (t-a_1)^{\alpha_1 - 1} \cdots (t-a_m)^{\alpha_m - 1} dt + B. \tag{3.5}$$




The Schwarz reflection principle shows that $g(z)$ reflects analytically across each interval on $\mathbb{R}$ between consecutive $a_j$'s, and also across the intervals $(-\infty, a_1)$ and $(a_m, +\infty)$. The reflected function across $(a_j, a_{j+1})$ maps the lower half-plane conformally onto the polygon obtained by reflecting $D$ across the straight line passing through $w_j$ and $w_{j+1}$. (The extensions obtained by reflecting across different intervals do not coincide.) Thus $g'(z)$ and $g''(z)$ extend analytically across each interval to the lower half-plane, and since $g(z)$ has no critical points on the intervals, the extended $g'(z)$ has no zeros. Thus $g''(z)/g'(z)$ extends analytically across each of the intervals to the lower half-plane. The form (3.1) of $g''(z)/g'(z)$ at the $a_j$'s shows that the reflections of $g''(z)/g'(z)$ across contiguous intervals coincide. Thus $g''(z)/g'(z)$ extends to be meromorphic on the entire complex plane, with a simple pole at each $a_j$ with residue $\alpha_j - 1$ and with no other poles. The theorem on corners (which applies to virtual corners) shows that $g''(z)/g'(z)$ is analytic at $\infty$ and vanishes there. Thus $g''(z)/g'(z)$ is a meromorphic function on the extended complex plane, hence rational, and (3.3) is its partial fractions decomposition (Section VI.4). Integrating (3.3), we obtain

$$\log g'(z) \;=\; \sum(\alpha_j - 1)\log(z - a_j) + \text{ constant},$$

and this yields (3.4) for some constant $A$. One more integration yields (3.5).

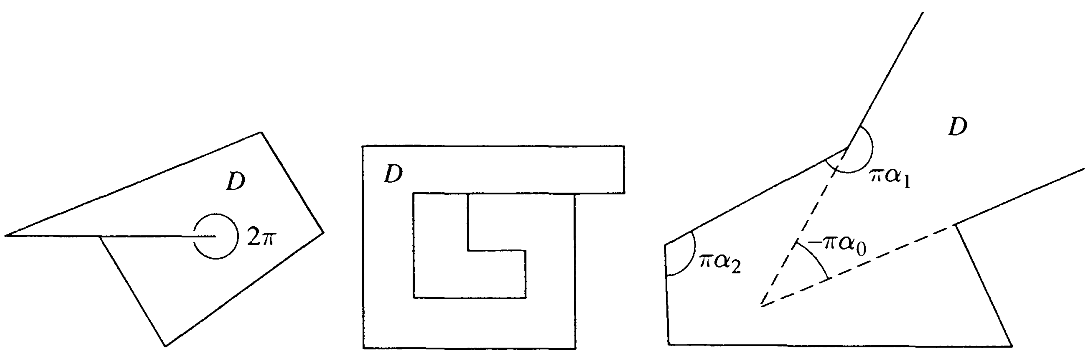
Both (3.4) and (3.5) are referred to as the Schwarz-Christoffel formula. In order to clarify the formula, we make a series of remarks.

1. Each of the singularities in (3.5) is integrable, since $\alpha_j > 0$.

2. If the angle at $w_j$ is $\pi$, so that we have only an apparent vertex, then $\alpha_j - 1 = 0$, and the corresponding factor disappears from the integral (as it should).

3. We can allow an angle of $2\pi$, corresponding to the tip of a slit. In this case we think of the slit as two intervals, one for each side of the slit.

4. We can allow intervals in $\partial D$ to overlap, and we can allow one point to serve as a multiple vertex, as in the figures.




5. We can allow $\infty$ to be a vertex, provided that we count the corresponding angle as negative. For instance, suppose $g(z)$ maps $a_1$ to the vertex $w_1 = \infty$, with angle $\pi\alpha_1$, where $-2 < \alpha_1 < 0$. The map to the $\zeta$-plane is then $\zeta(z) = g(z)^{1/\alpha_1} = (z - a_1)\psi(z)$, and we obtain $g(z) = (z - a_1)^{\alpha_1} h(z)$, where $h(z)$ is analytic at $a_1$ and $h(a_1) \neq 0$. We have (3.1) as before, and we obtain the same Schwarz-Christoffel formula (3.5). This time the factor $(z - a_1)^{\alpha_1}$ of $g'(z)$ is not integrable at $a_1$, corresponding to the fact that $g(a_1) = \infty$.

6. We can allow $\infty$ to be a vertex with angle zero. This corresponds to the "end" of an infinite strip. In this case the map to the $\zeta$-plane is given by $\zeta(z) = e^{\beta g(z)}$ for some $\beta$, and

$$g(z) \;=\; \frac{1}{\beta}\log\zeta \;=\; \frac{1}{\beta}\log(z - a_1) + h(z),$$

where $h(z)$ is analytic at $a_1$. A calculation reveals that (3.1) again holds. We are led to the same Schwarz-Christoffel formula (3.5), with $\alpha_1 = 0$ for the vertex at $\infty$.

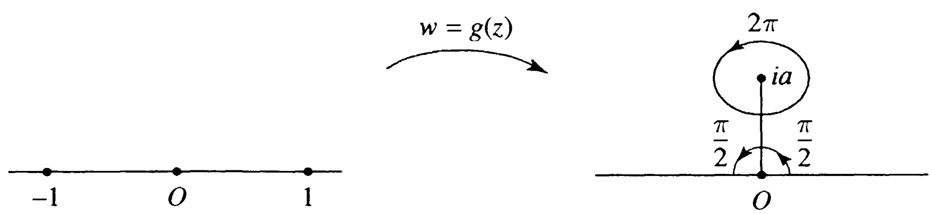
7. If $a_0 = \infty$ is mapped to a vertex $w_0 = \infty$, the Schwarz-Christoffel formula (3.5) holds, in which there is no factor corresponding to $\infty$. The

calculations are similar to those above. The result can also be obtained by letting the parameters in the Schwarz-Christoffel formula tend to $\infty$. (See Exercises 3 and 4.)

**Example.** Let $D$ be the domain obtained by cutting a vertical slit in the upper half-plane from 0 to $ia$ on the imaginary axis. (See Exercise 1.9.) Let $w = g(z)$ be the Schwarz-Christoffel map of the upper half-plane onto $D$ that sends 0 to $ia$ and $\pm 1$ to the two corners at 0. This determines $g(z)$ uniquely, since we have specified three real parameters. In the Schwarz-Christoffel formula, the consecutive points are $a_1 = -1$ with angle $\pi/2$, $a_2 = 0$ with angle $2\pi$, and $a_3 = +1$ with angle $\pi/2$. Thus (3.4) becomes

$$g'(z) \;=\; A(z-1)^{-1/2}z(z+1)^{-1/2} \;=\; Az/\sqrt{z^2-1}.$$

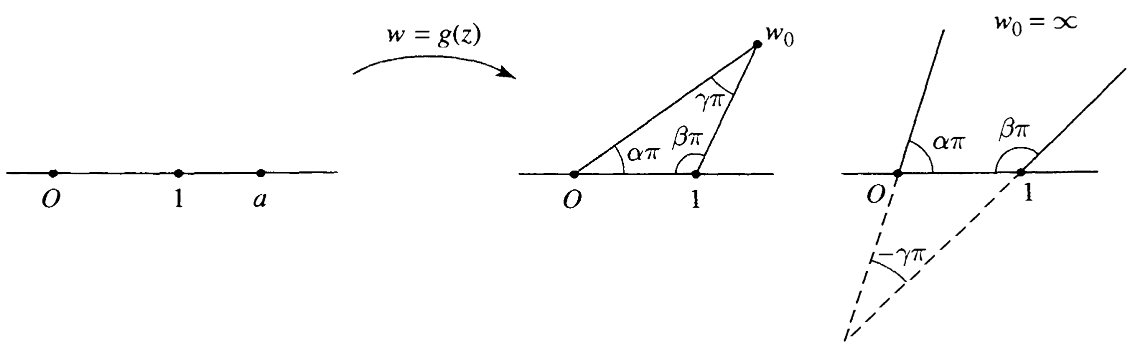
This can be integrated in closed form, and after matching boundary points to determine $A$ and the integration constant, we obtain $g(z) = a\sqrt{z^2-1}$, where the branch of the square root is the one that is positive on the interval $(1, \infty)$.




**Example.** Suppose $D$ is a triangle, with vertices at 0, 1, and a point $w_0$ in the upper half-plane, and interior angles $\alpha\pi$, $\beta\pi$, and $\gamma\pi$, where $\alpha + \beta + \gamma = 1$. Fix a point $a > 1$, and let $w = g(z)$ map the upper half-plane conformally onto $D$, with $g(0) = 0$, $g(1) = 1$, and $g(a) = w_0$. Since we have specified three real parameters, $g(z)$ is uniquely determined. Since $g(0) = 0$, the Schwarz-Christoffel formula becomes

$$g(z) \;=\; A\int_0^z t^{\alpha-1}(1-t)^{\beta-1}(a-t)^{\gamma-1}dt. \tag{3.6}$$

The constant $A$ is determined by the condition $g(1) = 1$.



## Exercises for XI.3

1. Derive the formula (3.2) for the case that $g(z)$ maps $\infty$ to the corner $w_0$ of $D$.

2. Let $\varphi(w)$ be the Riemann map of a simply connected domain $D$ onto $\mathbb{D}$. Suppose $w_0$ is a corner of $D$ with interior angle $\alpha\pi$. Set $w(\zeta) = w_0 + \lambda\zeta^\alpha$, where $|\lambda| = 1$. (a) Use the Schwarz reflection principle to show that $\varphi(w(\zeta))$ is analytic at $\zeta = 0$. (b) Show that $\varphi(w)$ has a series expansion $\sum b_j(w - w_0)^{j/\alpha}$ that converges uniformly for $w \in D$, $|w - w_0| < \varepsilon$. (c) Use (b) to show that $\varphi(w)$ has a limit as $w \in D$ tends to $w_0$.

3. Denote by $g_a(z)$ the conformal map of $\mathbb{H}$ onto the triangle with vertices 0, 1, and $w_0$ given by (3.6), so that $g_a(0) = 0$, $g_a(1) = 1$, and $g_a(a) = w_0$. Show that $g_a(z)$ converges as $a \to +\infty$, uniformly for $|z| \leq R$. Express the limit as an integral, and relate it to the Schwarz-Christoffel map.

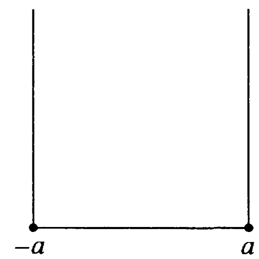
4. Fix $\alpha$ such that $0 < \alpha < 1$, and let $R > 0$. Denote by $f_R(z)$ the conformal map of $\mathbb{H}$ onto the triangle with vertices 0, 1, and $Re^{i\alpha\pi}$ given by (3.6), so that $f_R(0) = 0$, $f_R(1) = 1$, and $f_R(a) = Re^{i\alpha\pi}$. Show that $f_R(z)$ converges as $R \to \infty$, uniformly for $|z - a| \geq \varepsilon$. Express the limit as an integral, and relate it to the Schwarz-Christoffel map.

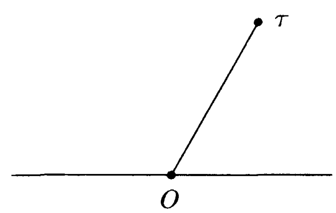
5. For fixed $0 < k < 1$, show that
$$w(z) = \int_0^z \frac{dt}{\sqrt{1-t^2}\sqrt{1-k^2t^2}}$$
maps the upper half-plane conformally onto a rectangle. Sketch the rectangle, using the notation
$$K = \int_0^1 \frac{dt}{\sqrt{1-t^2}\sqrt{1-k^2t^2}}, \quad K' = \int_1^{1/k} \frac{dt}{\sqrt{t^2-1}\sqrt{1-k^2t^2}}.$$
Show that the inverse $z = z(w)$ of the function defined above extends to be a meromorphic function $h(w)$ on the entire complex plane. Show that $h(w)$ is an odd function, and that $h(w)$ is doubly periodic with periods $4K$ and $2K'i$. Locate the zeros and the poles of $h(w)$, and indicate their orders. *Remark.* The integrals appearing above are **elliptic integrals**. The function $h(w)$ is the **Jacobian elliptic function** denoted classically by $\mathrm{sn}(w)$. It maps the torus formed by identifying the opposite edges of the period rectangle two-to-one onto the extended complex plane.

6. What is the Schwarz-Christoffel formula (3.4) for the conformal map $g(z)$ of the upper half-plane onto the horizontal strip $\{0 < \mathrm{Im}(w) <$

$a\}$ that maps 0 to $-\infty$ and $\infty$ to $+\infty$? By integrating, show that $g(z) = g(1) + \dfrac{a}{\pi} \operatorname{Log} z$.

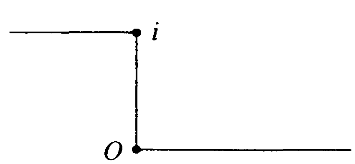
7. Use the Schwarz-Christoffel formula to find a conformal map $w = g(z)$ of the upper half-plane onto the vertical half-strip $\{\operatorname{Im} w > 0,\ -a < \operatorname{Re} w < a\}$ such that $g(1) = a$, $g(-1) = -a$, and $g(0) = 0$. Compare your result with that of Exercise 1.11.

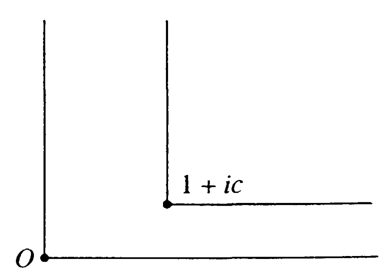



8. Let $\tau$ be a complex number such that $\operatorname{Im} \tau > 0$, and let $D_\tau$ be the domain obtained from the upper half-plane by deleting the straight line segment from 0 to $\tau$. (a) Find an integral expression for conformal maps $w = g(z)$ of the upper half-plane onto $D$ such that $g(0) = \tau$ and $g(\infty) = \infty$. (b) Show directly that if $C > 0$ and $0 < \sigma < 1$, the function $g(z) = C(z - \sigma)^\sigma (z + 1 - \sigma)^{1-\sigma}$ maps the upper half-plane conformally onto the domain $D_\tau$ for appropriate $\tau$. (c) How are the maps in (a) and (b) related?

9. Let $D$ be the step domain that is the union of the half-plane $\{\operatorname{Im} w > 1\}$ and the first quadrant $\{\operatorname{Re} w > 0, \operatorname{Im} w > 0\}$. Find a conformal map $w = g(z)$ of the upper half-plane onto $D$ satisfying $g(-1) = i$, $g(+1) = 0$, and $g(\infty) = \infty$.




10. For $c > 0$, find an integral expression for a conformal map $w = g(z)$ of the upper half-plane onto the corridor with a turn described by

$$D = \{0 < \operatorname{Re} w < 1, \operatorname{Im} w > 0\} \cup \{0 < \operatorname{Im} w < c, \operatorname{Re} w > 0\}.$$

In the case $c = 1$, find explicitly the conformal map $g(z)$ that satisfies $g(0) = 0$, $g(1) = +\infty$, and $g(\infty) = 1 + i$.

## 4. Return to Fluid Dynamics

We return to fluid dynamics, to see how conformal mapping techniques can be applied to solve fluid flow problems. Following the notation of Section III.6, we denote the complex velocity potential by $f(z) = \phi(z) + i\psi(z)$, where $\phi(z)$ is the velocity potential, and $\psi(z)$ is the stream function whose level sets are the flow lines of the fluid. The velocity vector field is $\mathbf{V}(z) = \overline{f'(z)}$. The flow is tangent to the boundary along boundary curves on which there are no sources or sinks. Thus $\psi(z)$ is a harmonic function on $D$ that is constant on each boundary curve with no sources or sinks.

**Example.** The stream function $\psi(z) = \operatorname{Im} z$ on the upper half-plane $\mathbb{H}$ corresponds to a flow parallel to the $x$-axis with no sources or sinks on the axis. The stream function $\psi(z) = \operatorname{Arg} z$ on $\mathbb{H}$ corresponds to a flow from a source at the origin. We may superimpose the flows, to obtain the stream function

$$\psi(z) \;=\; c_1 \operatorname{Im} z + c_2 \operatorname{Arg} z, \qquad \operatorname{Im} z > 0,$$

which corresponds to a flow near the bank of a large river with a pipe discharging an effluent at one point on the riverbank.

As in Section III.6, our strategy for solving a flow problem on a domain $D$ is to find a conformal map $h(w)$ from $D$ to the upper half of the $z$-plane, in order to transfer the boundary problem to $\mathbb{H}$. The transferred problem involves finding a harmonic function $\psi(z)$ on $\mathbb{H}$ that is constant on the intervals of the real axis corresponding to the boundary arcs of $D$ where there are no sinks or sources. The composed function $\psi(h(w))$ is then the stream function for the flow in $D$. In the case that $D$ is a polygon domain, the Schwarz-Christoffel formula gives the conformal map $w = g(z)$ from $\mathbb{H}$ onto $D$, and this must be solved for the inverse function $z = h(w)$ if a solution in closed form is required.

**Example.** Consider the flow near the bank of a large river with a small parallel tributary. We represent the configuration by the upper half of the $w$-plane with a horizontal slit terminating at the point $ib$ on the positive imaginary axis, $D = \mathbb{H}\backslash(ib - \infty, ib]$. Consider the conformal map $w = g(z)$ of $\mathbb{H}$ onto $D$ such that $-1$ corresponds to the tip of the slit, 0 to the source of the tributary, and $i\infty$ to $i\infty$. The three conditions $g(-1) = ib$, $g(0) = -\infty$, and $g(i\infty) = i\infty$ determine $g(z)$ uniquely. The function $g(z)$ maps the interval $(-\infty, -1]$ to the top edge of the slit, the interval $(-1, 0)$ to the bottom edge reversing direction, and the interval $(0, \infty)$ to the real axis $(-\infty, \infty)$. The Schwarz-Christoffel formula (4.3) becomes $g'(z) = A(z+1)z^{-1}$, corresponding to angle $2\pi$ at $-1$ and 0 at 0. The term involving $-\infty$ does not enter. Thus $g(z) = A(z + \operatorname{Log} z) + B$. The constants